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SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

SMART LIBRARY SYSTEM

A Capstone Project
Present to the Faculty of the
School of Information and Communication Technology
Passi City College
City of Passi, Iloilo

In Partial Fulfillment
of the Requirements for the Degree
Bachelor of Science in Information Technology

by:

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Porras, Chester J.



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APPROVAL SHEET

The Research/Capstone Project Study entitled "**SMART LIBRARY SYSTEM**", prepared and submitted by: **Wilson Jagolingay Jr., Jharaia Camille Labiaga, Krisha Lyne Lago, Russel Lasap Jr., and Chester Porras** in partial fulfillment of the requirements of the degree Bachelor of Science in Information Technology, has been examined and approved as requirements for GRADUATION.

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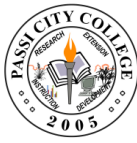
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GRAMMARIAN'S CERTIFICATE

This is to certify that the undersigned has reviewed and gone through all the proposed project pages entitled "**SMART LIBRARY SYTEM**" as against the structural rules that govern the composition of sentences, phrases, and words in the English language.

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DEDICATION

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SMART LIBRARY SYSTEM MAINTENANCE

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ABSTRACT

As digital technology continued to advance, traditional library systems that relied on manual processes often experienced problems such as slow transactions, repeated data entry, and limited access to real-time information. To address these issues, the project presented a Smart Library System designed to improve library operations through automation and digital access. The system consisted of two main interfaces: a User Portal and an Administrative Dashboard. The User Portal allowed students and faculty to search the library catalog, view book details, and check availability online without the need to visit the library. The Administrative Dashboard helped library staff manage borrowing and returning processes using barcode scanning and student identification, making transactions faster and more accurate. It also enabled administrators to manage student records and book inventories, generate reports, and send announcements to users. Overall, the Smart Library System aimed to reduce manual work, minimize errors, and create a more efficient and user-friendly library environment.

Keywords: *Smart Library System, User Portal, Administrative Dashboard, Waterfall Methodology*



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CHAPTER I

INTRODUCTION

Project Context

Background of the Study

Libraries played a crucial role in academic institutions by providing students and faculty with access to a wide range of learning resources. However, traditional library systems often faced challenges such as manual book searches, inefficient borrowing processes, and limited accessibility. These inefficiencies frequently led to delays, misplaced resources, and reduced user satisfaction.

The Passi City College Library previously relied on conventional library management practices, which did not fully meet the demands of a growing learner population. The absence of a digital catalog and an automated book-tracking system resulted in difficulties for users in locating available books and managing their borrowing history. To address these concerns, the integration of a Smart Library System with an Online Public Access Catalog (OPAC) was proposed.

A Smart Library System enhanced library operations through automation and digitalization. It incorporated features such as barcode-based book tracking, real-time



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availability status, self-service borrowing and returning, and automated fine collection. The integration of an OPAC allowed users to search for books and check availability efficiently. This digital transformation aligned with modern library standards and significantly improved accessibility, operational efficiency, and user engagement.

By implementing this system, Passi City College aimed to modernize its library services and ensure a seamless, technology-driven experience for students and faculty. The study focused on the design, development, and evaluation of the proposed Smart Library System with OPAC, assessing its impact on library efficiency and user satisfaction.

Statement of the Problem

The Passi City College Library previously operated using traditional methods for book cataloguing, borrowing, and returning, which were time-consuming and inefficient. Students and faculty members experienced difficulties in searching for books, checking availability, and managing borrowing records due to the lack of an automated system. Moreover, manual record-keeping increased the risk of errors, misplacements, and delays in updating book statuses.



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This study aimed to develop and integrate a Smart Library System with an Online Public Access Catalog (OPAC) to enhance the efficiency, accessibility, and user experience of library services. Specifically, it sought to answer the following research questions:

Research Questions:

What challenges were encountered by students and faculty in accessing and managing library resources at Passi City College?

How did the integration of a Smart Library System with OPAC improve book searching, borrowing, and returning processes?

Project Description

The Smart Library System with Online Public Access Catalog (OPAC) for the Passi City College Library was a digital library management solution designed to modernize library services. The system aimed to enhance book accessibility, automate borrowing and returning processes, and improve overall library efficiency through advanced technologies.

Through OPAC integration, students were able to search for books and check real-time availability, while faculty and



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administrators could add books and users, generate reports, notify students about overdue materials, and record attendance. The system incorporated automated borrowing and returning using barcode technology, real-time book tracking, and a user-friendly interface.

An integrated penalty and payment system ensured proper fine management for overdue books, offering convenient digital payment options. A dedicated dashboard enabled library staff to manage inventories, monitor borrowing activities, and generate reports efficiently.

The implementation of the Smart Library System significantly enhanced library service efficiency by reducing manual workloads, minimizing errors, and improving access to resources. The system supported Passi City College's goal of embracing digital transformation in education by providing a secure, accessible, and technology-driven library environment.

Objectives of the Study

The Smart Library System integrated with an Online Public Access Catalog (OPAC) for the Passi City College Library aimed to develop and implement a modernized library service.



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General Objective:

The general objective of the study was to develop a Smart Library System with OPAC that enhanced the efficiency and accessibility of the Passi City College Library. The system automated book searching, borrowing, and returning processes, provided real-time availability updates, and integrated fine notifications. The study also evaluated the effectiveness of the system in streamlining library operations and improving user satisfaction.

Specific Objectives:

- Automated book borrowing and returning processes.
- Tracked real-time book availability and inventory.
- Integrated an Online Public Access Catalog (OPAC) for efficient book searching.
- Provided a dedicated dashboard for library staff to manage inventories, user records, and reports.
- Incorporated barcode scanning technology for efficient book identification and tracking.



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Scope and Delimitations of the Study

The study focused on the development and implementation of a Smart Library System integrated with OPAC for the Passi City College Library. It covered automated book searching, borrowing, returning, and inventory management.

The system included real-time availability tracking, administrator account management, overdue fine processing, automated notifications, barcode scanning technology, and a staff dashboard. It was designed to be mobile-responsive and incorporated data security measures to protect user information.

However, the study was limited to the library operations of Passi City College and did not extend to other institutions. It focused solely on physical book management and excluded digital libraries and e-book integration. Testing was conducted in a controlled environment, and full-scale deployment would require additional resources.

External factors such as network instability and server downtime could temporarily affect system availability and were beyond the study's control. While the system minimized human error, its accuracy depended on proper usage and user training. Future enhancements could include e-book



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integration, AI-based search recommendations, and cloud-based storage solutions.

Significance of the Study

The Smart Library System with OPAC was significant in improving library operations, accessibility, and user experience.

Students. The system provided students with convenient access to library resources, enabling them to search books, check availability, reserve materials, and manage borrowing records online.

Faculty. The system simplified access to teaching and research materials, supporting academic instruction and scholarly activities.

Library Staff. Automation improved efficiency by reducing manual tasks such as borrowing, returning, inventory tracking, and fine collection, while barcode scanning minimized errors and processing time.

Passi City College. The implementation supported the institution's digital transformation goals and enhanced its reputation as a technologically progressive academic environment.



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Future Researchers and Developers. The study served as a reference for similar digital library projects, offering insights into system design, implementation challenges, and possible improvements.

Definition of Terms

Barcode. A machine-readable representation of data consisting of parallel bars and spaces used to identify specific items.

Database. An organized collection of structured data stored electronically and managed by a database management system.

Firestore. A Google-developed platform providing tools and services for building and managing web and mobile applications.

Library System. A manual or computerized system used to manage library operations such as cataloguing, circulation, and resource management.

OPAC. Online Public Access Catalog; a digital database that allows users to search and locate library materials.

XAMPP. An open-source web server solution stack that includes Apache, MariaDB, PHP, and Perl.



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CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter discussed the necessity of developing a Smart Library System integrated with an Online Public Access Catalog (OPAC) for the Passi City College Library. The review highlighted the significant role of a technology-driven library system in enhancing book management, improving transaction efficiency, and increasing the accessibility of library resources for students, faculty, and staff. This initiative is supported by Republic Act No. 9246, also known as the Philippine Librarianship Act of 2003, which recognizes the importance of modern and computerized library systems in providing efficient information services and promoting intellectual development.

The reviewed literature emphasized the significance of implementing a Smart Library System integrated with OPAC to enhance library operations, improve resource accessibility, and streamline inventory management. Various studies highlighted the benefits of automated cataloging, real-time book tracking, and online reservation systems, which reduced manual workload and improved service efficiency. The integration of RFID or barcode technology was found to minimize errors in book lending and returning, thereby



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ensuring accurate record-keeping and reducing the incidence of misplaced materials. These findings align with RA 9246, which encourages the use of information technology and digital tools in library operations to improve service delivery and information retrieval.

Additionally, the literature underscored the importance of a centralized digital platform for managing borrowing transactions, issuing overdue notifications, and generating reports, which led to a more structured and organized library system. By adopting an advanced library management solution, institutions such as Passi City College were able to comply with the standards set by the Philippine Librarianship Act, which promotes professional, efficient, and technology-assisted library services. This adoption resulted in a more efficient, user-friendly, and technology-driven library experience, ultimately enhancing academic support services for students and faculty.

In summary, the review of related literature strongly supported the need for a Smart Library System with an Online Public Access Catalog (OPAC) for Passi City College. The studies reviewed highlighted the importance of maintaining organized and updated library records, ensuring efficient book tracking, and utilizing a centralized system for managing



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library operations, as mandated by Republic Act No. 9246. A well-developed system was shown to facilitate smooth transactions, minimize errors, and enhance accessibility for students and faculty. By adopting a reliable and technology-driven library system, Passi City College could improve its library services and academic support, contributing to the institution's overall growth and success.

Local Review of Related Literature

Knowledge, satisfaction, and challenges on Online Public Access Catalog in an academic library in Northern Philippines (2020). Examined undergraduate students' knowledge, satisfaction, and challenges in using the Online Public Access Catalog (OPAC) at Saint Mary's University in Northern Philippines.

Their findings showed that students had limited understanding of OPAC's functions, reported only moderate satisfaction, and experienced several difficulties, with notable variations based on gender, school affiliation, and frequency of use. The researchers stressed the need to improve OPAC services to enhance accessibility and meet users' needs. In contrast, a Smart Library System not only incorporates OPAC but also extends its capabilities by adding automated



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circulation, inventory management, user accounts, notifications, and RFID/barcode features, thereby addressing OPAC's limitations and offering a more user-friendly, comprehensive, and efficient platform for academic libraries.

Automated library management system for public libraries in the Philippines (2021). Developed an automated library management system for public libraries in the Philippines to enhance efficiency, oversight, and user accessibility. Built using PHP, JavaScript, HTML, CSS, and SQL under the agile methodology, the system achieved an overall evaluation rating of 4.42 or "Excellent," reflecting high user satisfaction.

The study emphasized the importance of dependable and adaptable systems, particularly during crises like pandemics when physical access is restricted. Similarly, the Smart Library System also focuses on automating library processes but is tailored for academic institutions, offering features such as OPAC integration, circulation management, inventory control, user accounts, notifications, and RFID/barcode support. While Obsanga and Enierga concentrated on strengthening public libraries for resilience in emergencies, the Smart Library System expands this concept by addressing



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the broader academic needs of students and faculty, showcasing the growing shift toward digital automation to improve efficiency, accessibility, and adaptability across different library settings.

Students' awareness and use of the Online Public Access Catalog (OPAC) at the Central Luzon State University Library in the Philippines (2024). Investigated students' awareness and use of the Online Public Access Catalog (OPAC) at the Central Luzon State University Library, recognizing its role as a digital gateway to academic resources.

The study found that although OPAC was widely utilized, many students were unaware of its full features, with limited orientation programs and insufficient terminals identified as key barriers. While most students were satisfied, some expressed dissatisfaction due to these limitations. The research highlighted the need to improve OPAC systems and instructional services to better match students' technological and academic needs. In comparison, the Smart Library System not only integrates OPAC but also extends its functions through automated circulation, inventory management, user accounts, notifications, and RFID or barcode technologies, offering a more comprehensive and accessible



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solution for academic institutions. Both works emphasize the importance of modernizing library services to meet students' evolving expectations in the digital age.

Foreign Review of Related Literature

Online Public Access Catalogue (OPAC) users in academic libraries: A case study (2020). Conducted a case study at Madurai Kamaraj University Library to investigate the role of the Online Public Access Catalogue (OPAC) in academic libraries, focusing on its function as a bibliographic search and retrieval tool. Using a questionnaire survey of 150 participants, with 132 valid responses, the study analyzed user experiences, usage patterns, and functional issues, revealing both the advantages of OPAC and the areas that require improvement.

While the research primarily provided insights and recommendations based on user perceptions, the Smart Library System offers a broader and more advanced approach by integrating OPAC with automated services such as circulation management, inventory control, user accounts, notifications, RFID, and cloud hosting. Unlike OPAC, which serves mainly as a search interface for students and staff, the Smart Library System delivers a comprehensive prototype designed for



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students, faculty, librarians, and administrators, aiming to enhance efficiency, accessibility, and service delivery in academic libraries. Both highlight the importance of improving library systems, but the Smart Library System provides a holistic, technology-driven solution that addresses limitations identified in OPAC.

Development of an open-source automated library system with Book Recommendation System for Small Libraries (2020).

Developed an open-source automated library system for small libraries that uniquely integrates a machine learning-based book recommendation feature. Using support-vector machines, the system analyzed bibliographic records, title similarity, classification, and borrowing history to generate personalized book suggestions, with evaluation conducted at Banpasao Chiangmai School using over 2,500 books and 4,600 loan records. Designed for accessibility, it allowed librarians to freely download and utilize the OPAC-based platform for catalog management and recommendation services. In contrast, the Smart Library System also automates library processes and incorporates OPAC but expands its scope by including circulation management, inventory control, user accounts, notifications, and RFID/barcode features for



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academic institutions. While Puritat and Intawon emphasized personalization and intelligent recommendations for small-scale libraries, the Smart Library System provides a broader, integrated platform to enhance efficiency and accessibility in academic settings.

Online Public Access Catalogue (OPAC) among library users: A case study of Olusegun Oke Library, LAUTECH, Ogbomoso, Oyo State, Nigeria (2021). Investigated the use of the Online Public Access Catalogue (OPAC) at the Olusegun Oke Library, LAUTECH, analyzing its purpose, frequency of use, user satisfaction, and challenges through a descriptive survey of 130 respondents. The study found that OPAC was commonly used to locate journals and check book availability, but users faced significant issues such as frequent system malfunctions, slow searches, high failure rates, and insufficient staff guidance.

To improve utilization, the authors recommended enhancing system functionality and providing better user support. In comparison, the Smart Library System builds on OPAC by integrating advanced features like circulation management, inventory tracking, user accounts, notifications, and RFID/barcode support, offering faster, more reliable, and



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user-friendly services. While Adegun et al. emphasized OPAC's operational limitations and the need for improvement, the Smart Library System delivers a comprehensive technological solution aimed at minimizing errors and enhancing efficiency for students, faculty, and administrators.

Online public access catalogues and library discovery systems (2022). Explored the transformation of Online Public Access Catalogues (OPACs) into modern discovery systems, tracing their origins from the automation era of the 1960s to their present role in digital information retrieval. The article examined how technical standards, along with usability, semiotic, and ethical challenges, have shaped their development and effectiveness over time. It also raised concerns about the growing dominance of internet search engines, questioning how libraries can adapt to shifting user behaviors and expectations.

In contrast, the Smart Library System builds upon OPAC's foundation but expands its scope through features like automated circulation, inventory management, user accounts, notifications, RFID, and cloud hosting, offering a practical solution tailored for academic libraries. While Wells provided a broad historical and conceptual view of OPACs and



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discovery systems in the evolving information landscape, the Smart Library System represents a localized, technology-driven implementation aimed at improving accessibility, efficiency, and usability. Both highlight the need for library systems to evolve in order to remain effective and relevant in the digital age.

Utilization of Online Public Access Catalogue (OPAC) as a retrieval tool for information resources in some selected university libraries in South-South Nigeria (2023).

Investigated the use of OPAC as an information retrieval tool in selected university libraries in South-South Nigeria, showing that it allows users to search by title, author, keyword, and other bibliographic details while enhancing record retrieval and presenting results in a user-friendly format. Usage patterns varied from daily to monthly, particularly during examination periods. However, challenges such as limited user knowledge, unstable electricity, insufficient computer terminals, and navigation difficulties hindered OPAC's effectiveness.

The researchers recommended continuous orientation programs, infrastructure investment, and policies promoting OPAC use to improve efficiency. In contrast, the Smart Library



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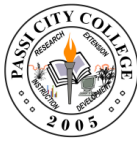
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System integrates OPAC with broader features like circulation management, inventory control, user accounts, notifications, and RFID/barcode tracking, addressing these limitations through automation and technological support.

While Amadi et al. focused on user behaviors and infrastructural challenges, the Smart Library System provides a comprehensive, technology-driven solution to enhance reliability, usability, and overall library services.

Access to information resources using the Online Public Access Catalogue (OPAC) in the 21st century (2024). Explored the role of OPAC in accessing information resources in modern libraries, contrasting it with traditional manual catalog systems and emphasizing its growing importance in the digital era. The study highlighted the need for skilled catalogers and resource classifiers in managing digital cataloging while also identifying challenges such as technical issues, lack of user awareness, and inadequate training that hinder effective OPAC use.

To improve accessibility and efficiency, the authors recommended promoting OPAC utilization through user education and information literacy programs. In contrast, the Smart Library System builds on OPAC by incorporating advanced



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features like automated circulation, inventory tracking, user accounts, notifications, and RFID/barcode technologies, offering a more seamless and efficient library experience. Whereas Nwobu and George focused on competencies and user literacy as key factors in OPAC effectiveness, the Smart Library System addresses these concerns with a holistic, technology-driven approach that minimizes technical barriers and enhances usability in academic libraries.



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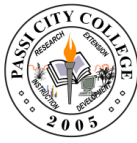
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CHAPTER III

METHODOLOGY

This study utilized a descriptive-developmental research design. The descriptive component aimed to provide an accurate and systematic analysis of the existing condition and needs of the Passi City College Library, particularly in terms of book inventory management, borrowing transactions, and the accessibility of library resources. Data were gathered through surveys, interviews, and observations to identify existing challenges, inefficiencies, and user requirements related to library operations.

The developmental component focused on the design, development, evaluation, and implementation of a Smart Library System integrated with an Online Public Access Catalog (OPAC) for the Passi City College Library. This phase involved the creation of a web-based system that automated book searching, reservations, borrowing, and returning processes while integrating barcode or RFID technology for efficient resource tracking. The system was developed using modern programming languages and development tools to ensure accessibility, security, and ease of use for students, faculty, and library staff.



Project Development

System Development Process Model

The researchers applied the principles of the System Development Life Cycle (SDLC). Specifically, the SDLC Waterfall Model was utilized to guide the systematic development of the proposed system.

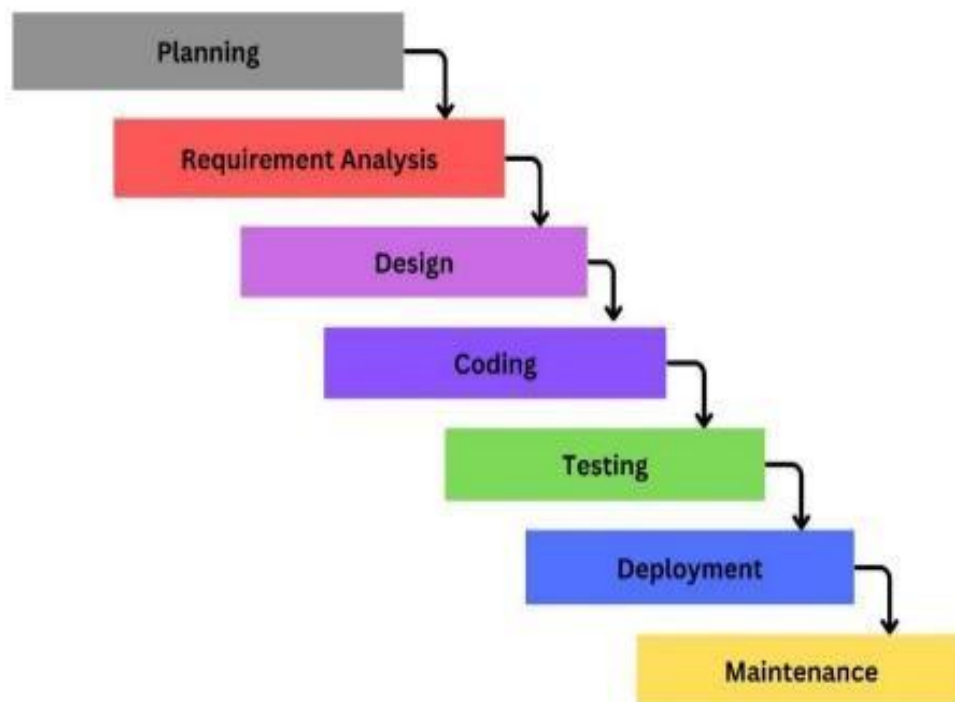
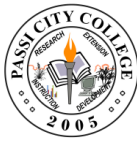


Figure 1. SDLC Waterfall Methodology Model

Figure 1 illustrates the Waterfall Development Methodology used in creating the system. This study utilized the System Development Life Cycle (SDLC) Waterfall Model developed by Pressman (2010). The model is composed of five



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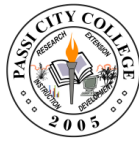
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phases: Planning, Requirement Analysis, Design, Development, Testing, and Maintenance.

Planning. In the planning phase, the project team conducted a feasibility study to determine whether the Smart Library System with an Online Public Access Catalog (OPAC) was technically feasible, financially viable, and operationally sustainable. This phase involved assessing the project's potential benefits, required resources, and possible challenges to ensure successful implementation.

Requirement Analysis. During this phase, the requirements identified in the planning stage were analyzed in detail. The team gathered input from librarians, students, and faculty to identify essential system functionalities such as cataloging, search features, borrowing and returning processes, and user access levels. This phase also involved the preparation of detailed functional specifications and design documents outlining the system's capabilities and constraints.

Design. The design phase involved the creation of a blueprint for the Smart Library System with OPAC. A high-level system architecture was developed, followed by a detailed design specifying system components and modules. This phase included the design of the database structure,



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user interface wireframes, and system workflows to optimize efficiency and enhance user experience.

Design for user interface.

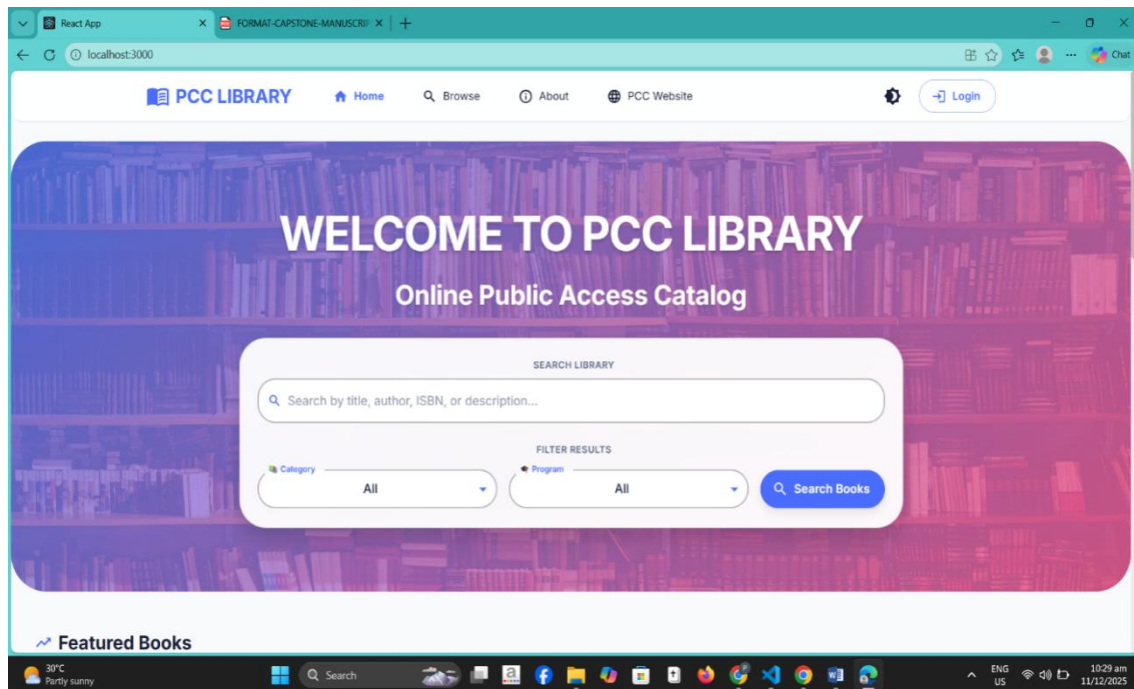


Figure 2. Design for User Interface



Design for Login Page.

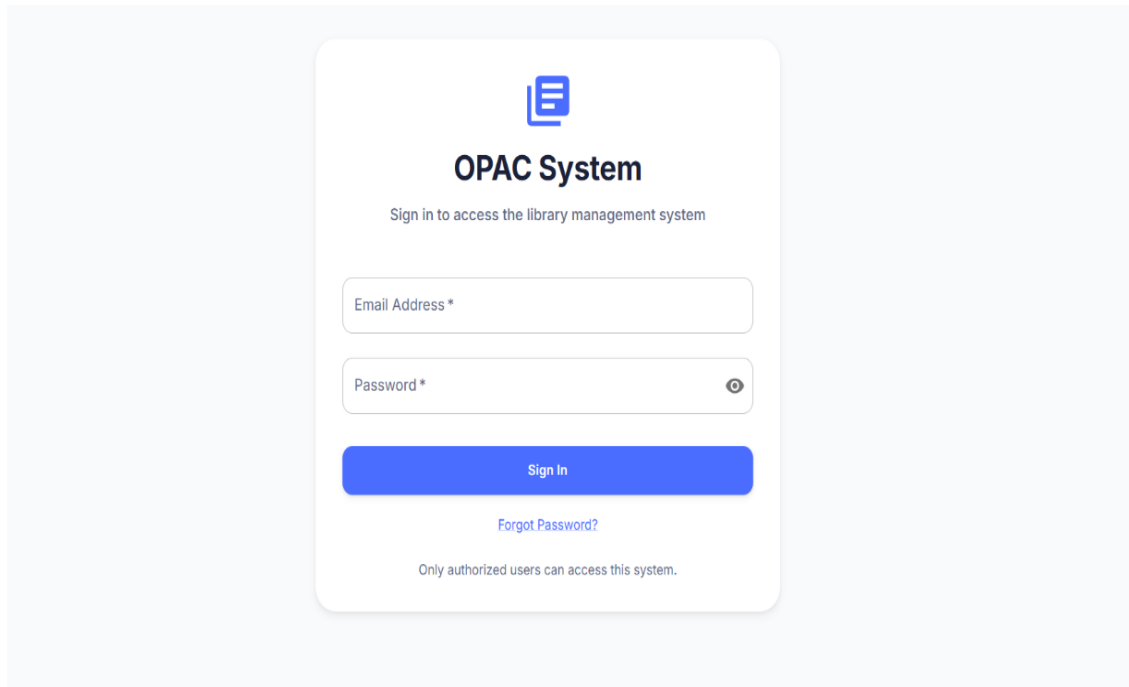


Figure 3. Design for Login Page

Design for Dashboard

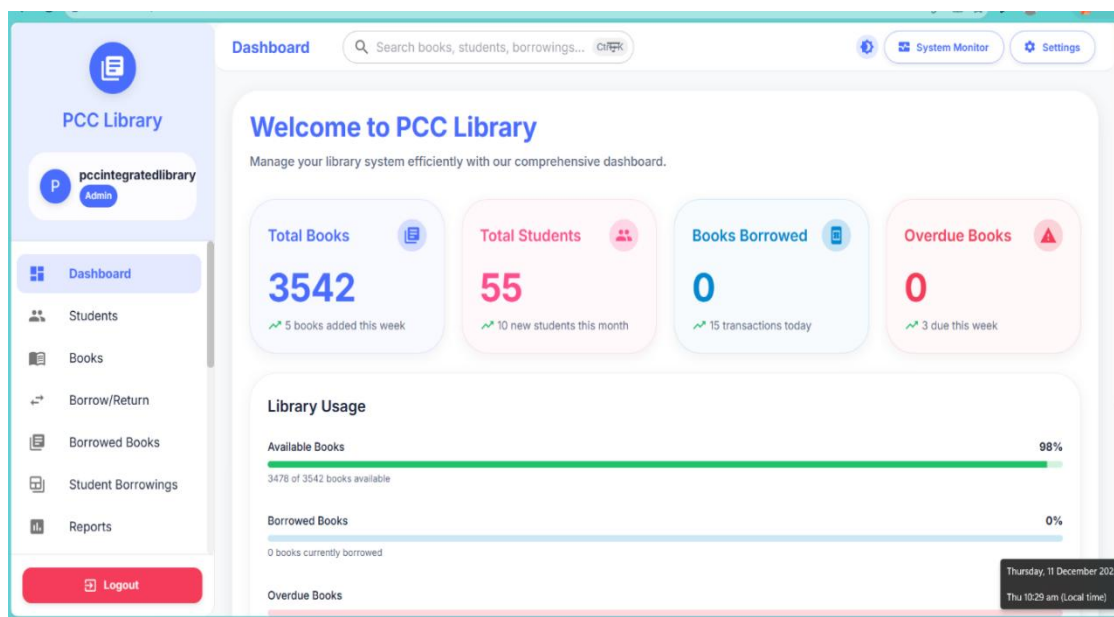


Figure 4. Design for Dashboard



Design for Student Management.

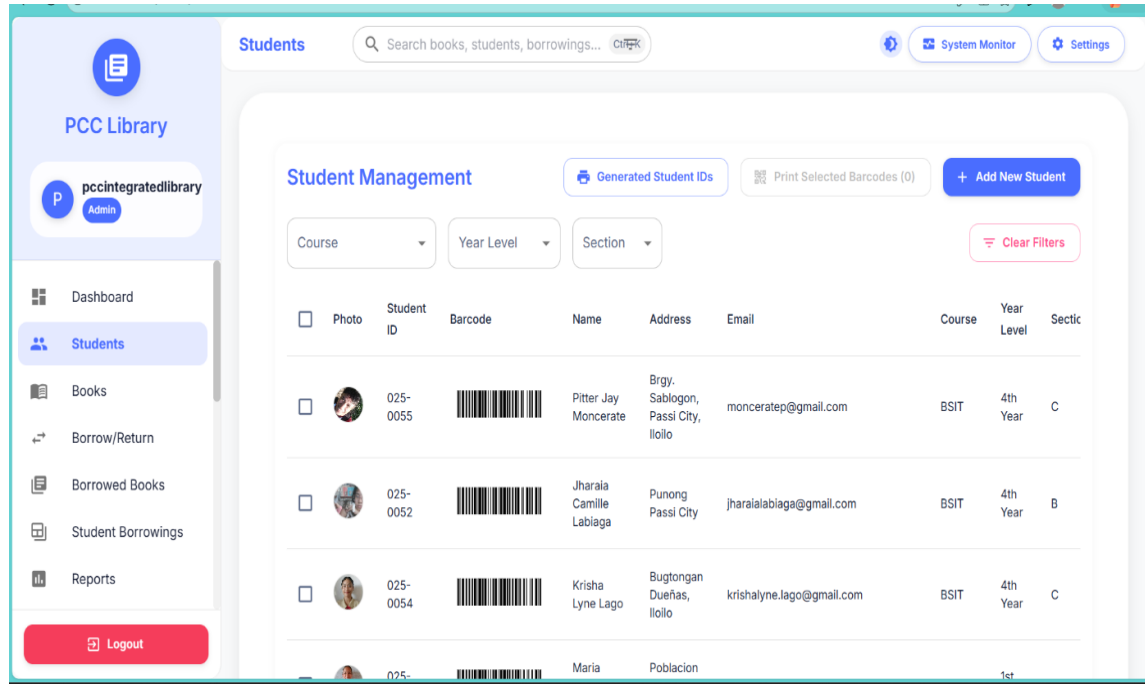


Figure 5. Design for Student Management

Design for Book Management

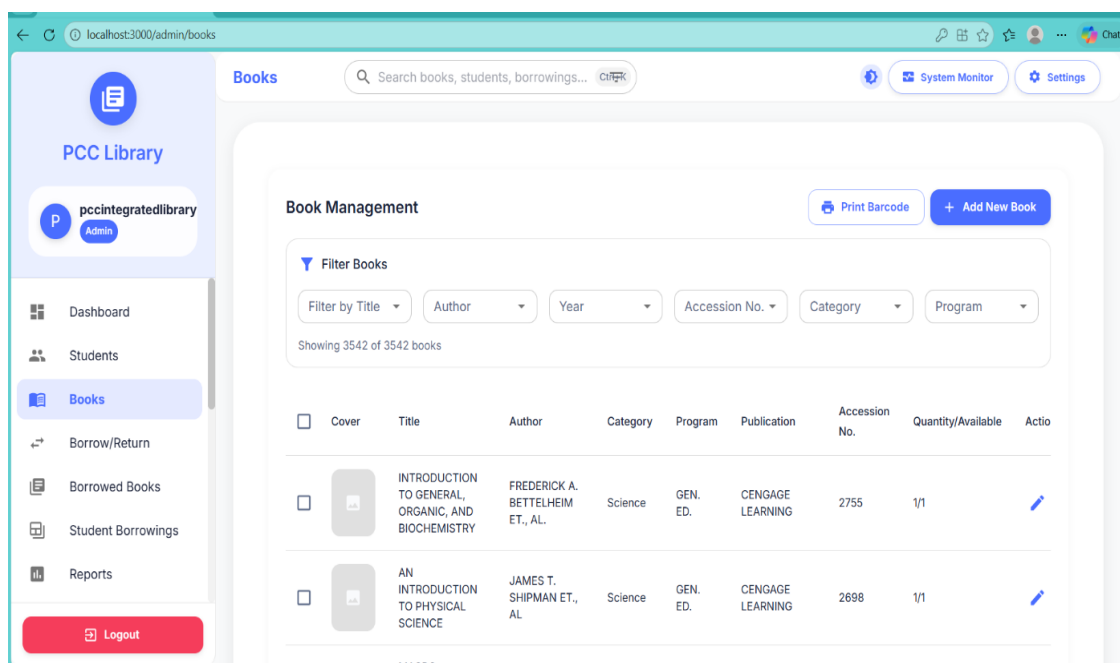


Figure 6. Design for Book Management



Design for Borrow/Return.

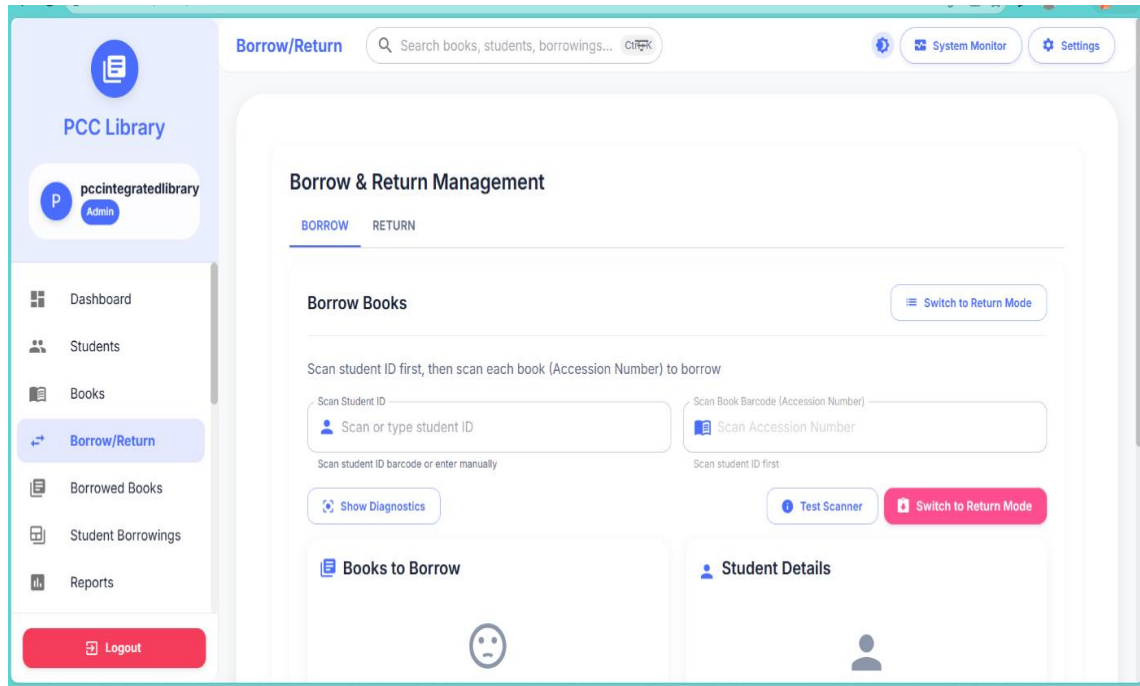


Figure 7. Design for Borrow/Return

Design for Borrowed Books.

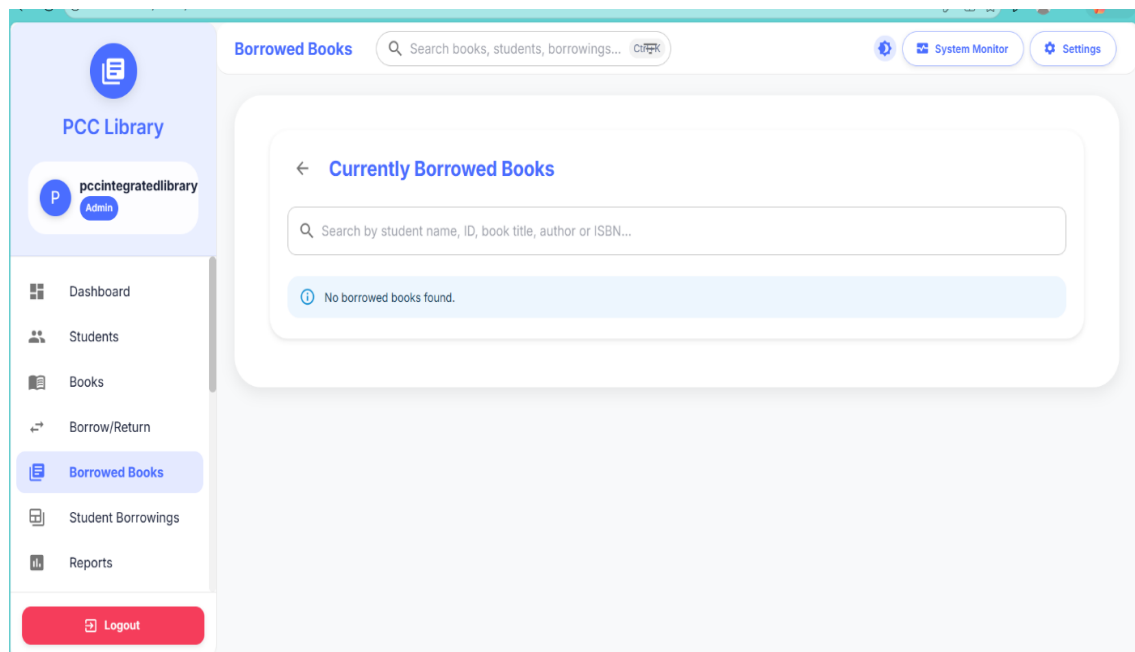


Figure 8. Design for Borrowed Books



Design for Student Borrowings.

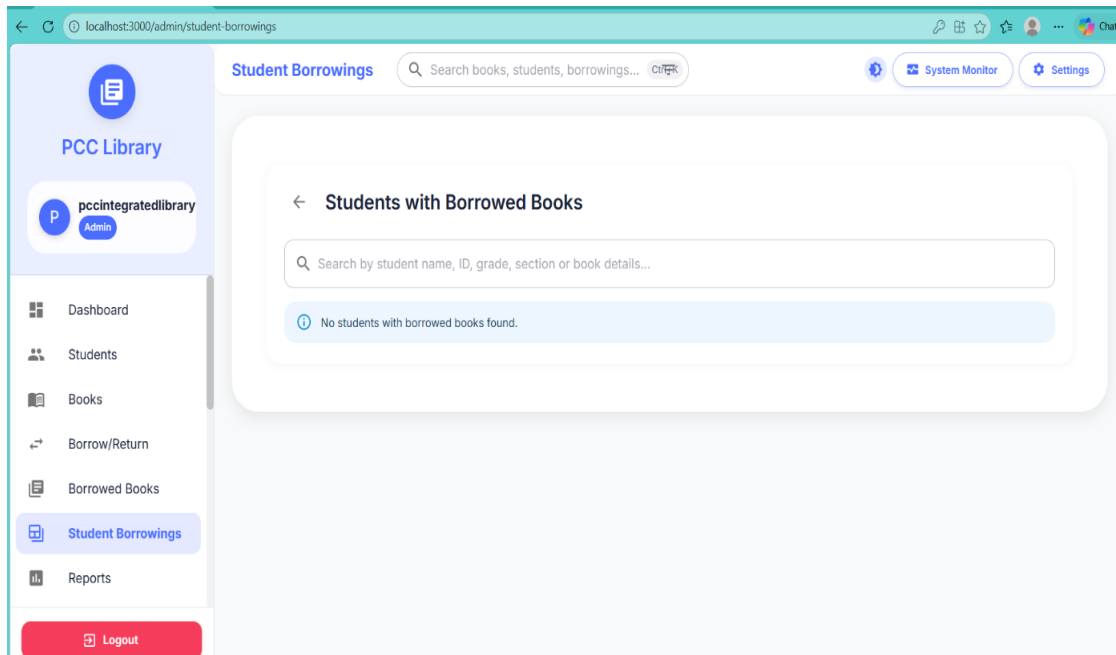


Figure 9. Design for Student Borrowings

Design for Reports.

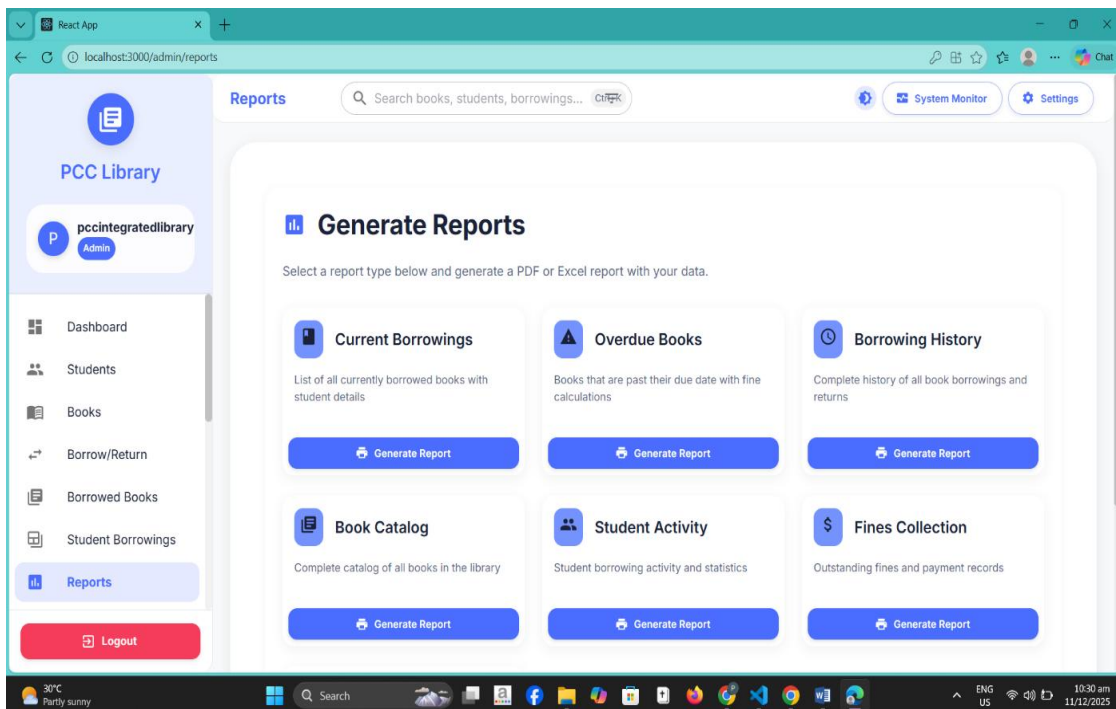


Figure 10. Design for Reports



Design for Announcements.

The screenshot displays the 'Send Announcements' form within the PCC Library Admin interface. The left sidebar contains navigation links: Borrow/Return, Borrowed Books, Student Borrowings, Reports, Announcements (selected), Overdue Notifications, Fine Payments, and Attendance. The main content area has a search bar and buttons for System Monitor and Settings. The form itself includes fields for Subject, Message, and an optional Attachment URL. A 'Send to 55 Filtered Students' button is at the bottom right. The browser address bar shows 'localhost:3000/admin/announcements'.

Figure 11. Design for Announcements

Design for Overdue Notifications.

The screenshot displays the 'Overdue Books Notifications' section of the PCC Library Admin interface. The left sidebar is identical to the previous figure, with 'Overdue Notifications' selected. The main content area shows a search bar, System Monitor, and Settings buttons. Below these is a 'Refresh' button and a message stating 'No overdue books found'. A 'Create Test Overdue Data' button is also present. The browser address bar shows 'localhost:3000/admin/overdue-notifications'.

Figure 12. Design for Overdue Notifications



Design for Fine Payments.

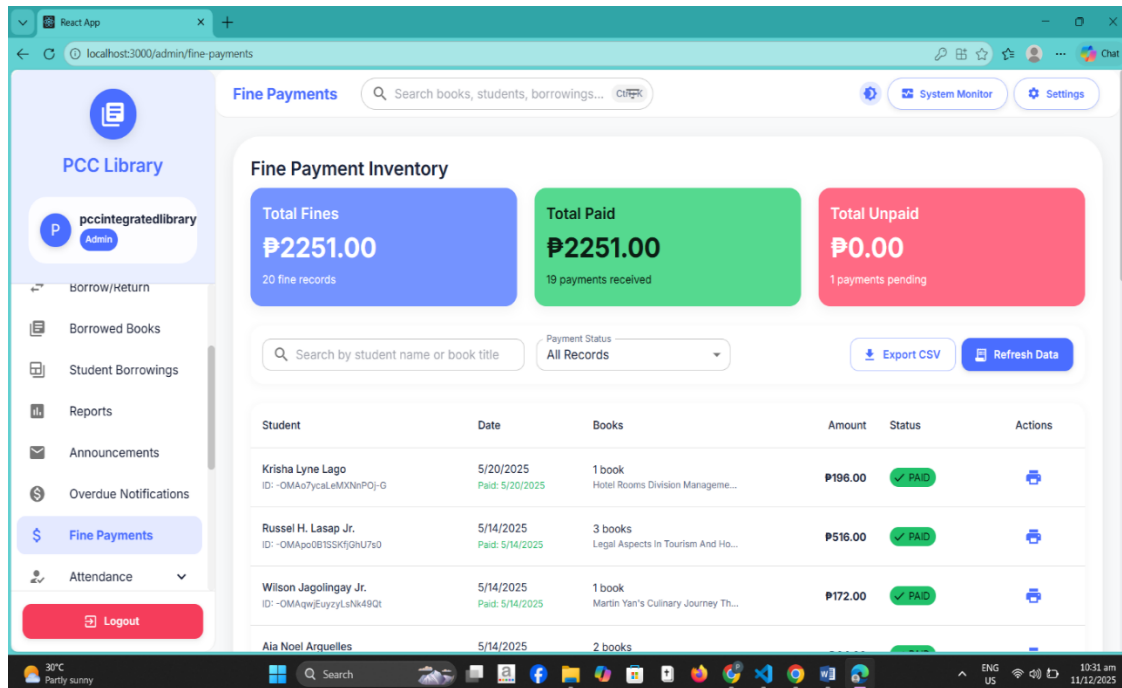


Figure 13. Design for Fine Payments

Design for Record Attendance.

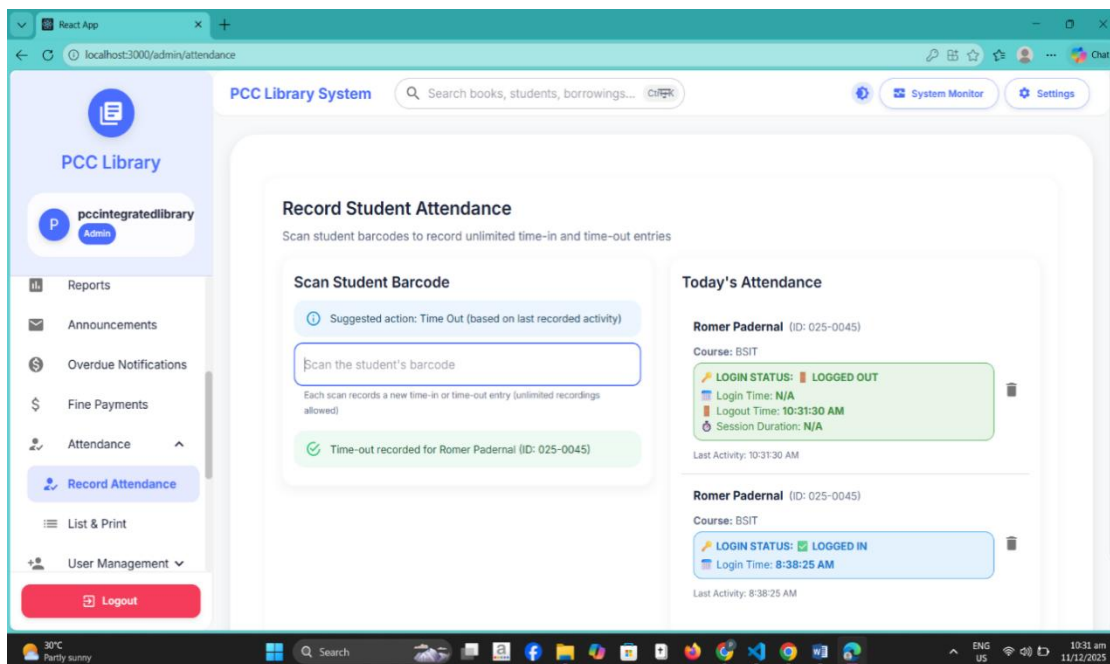


Figure 14. Design for Record Attendance



Design for List & Print Attendance.

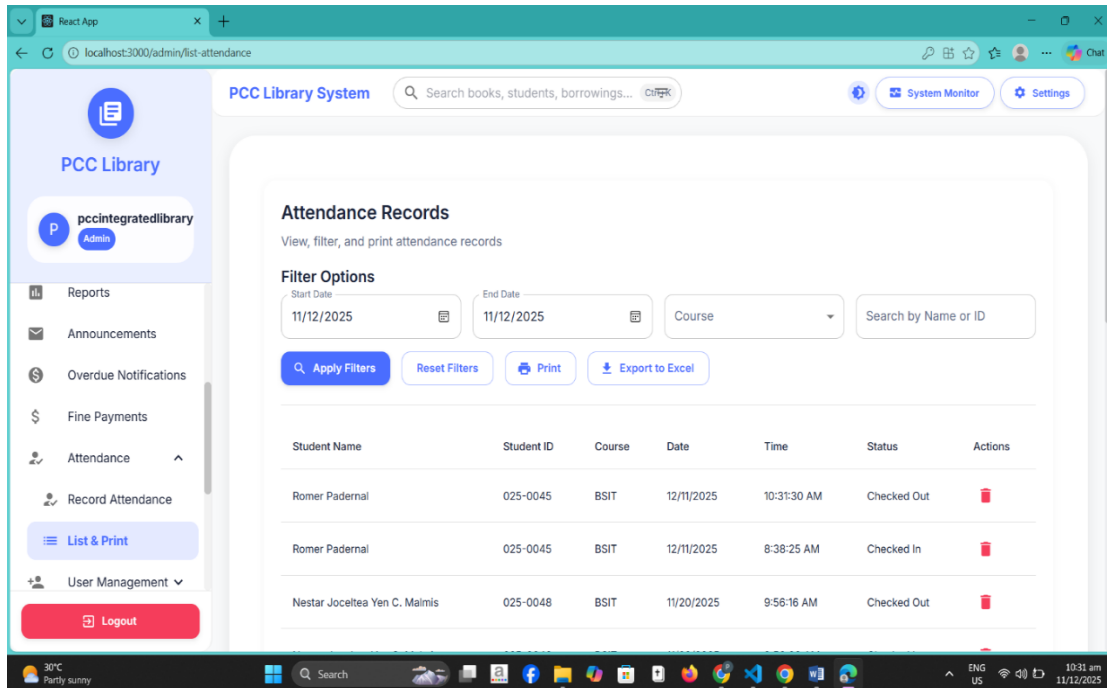


Figure 15. Design for List & Print Attendance

Design for User Management.

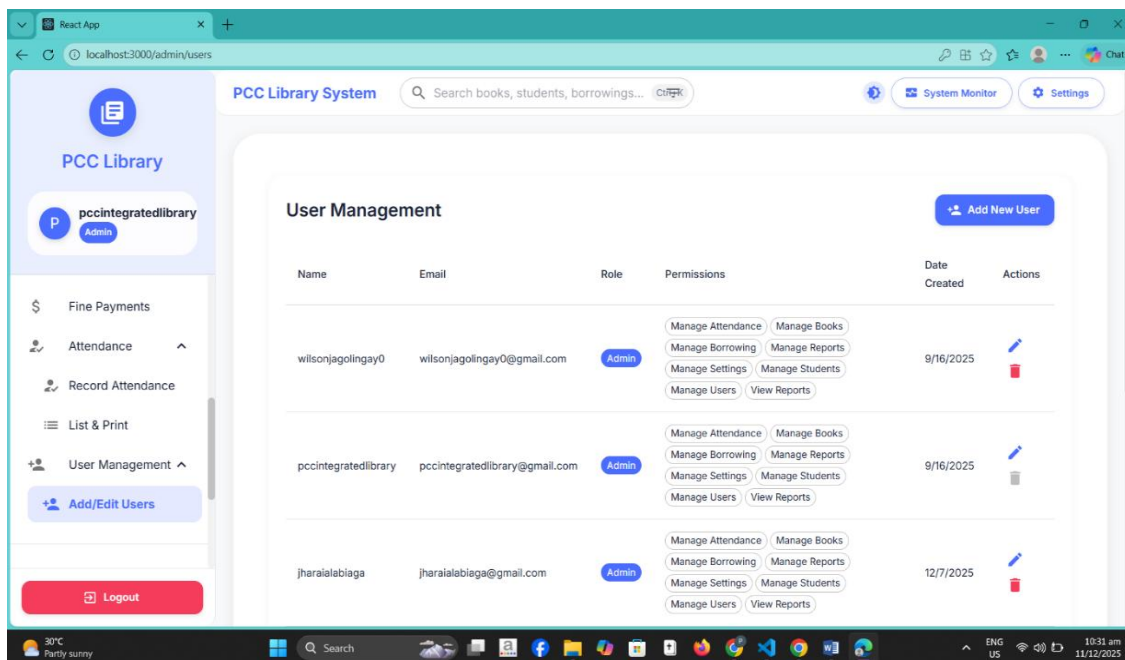


Figure 16. Design for User Management



Design for System Monitor.

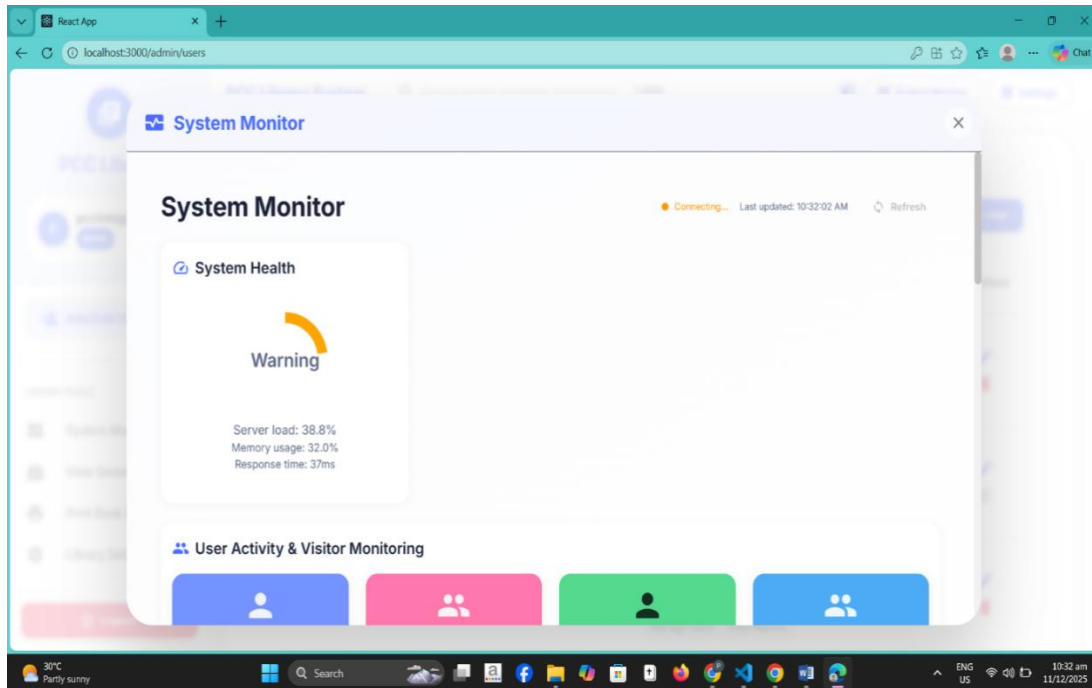


Figure 17. Design for System Monitor

Design for View Generated Ids.



Figure 18. Design for View Generated IDs



Design for Print Book Barcode.

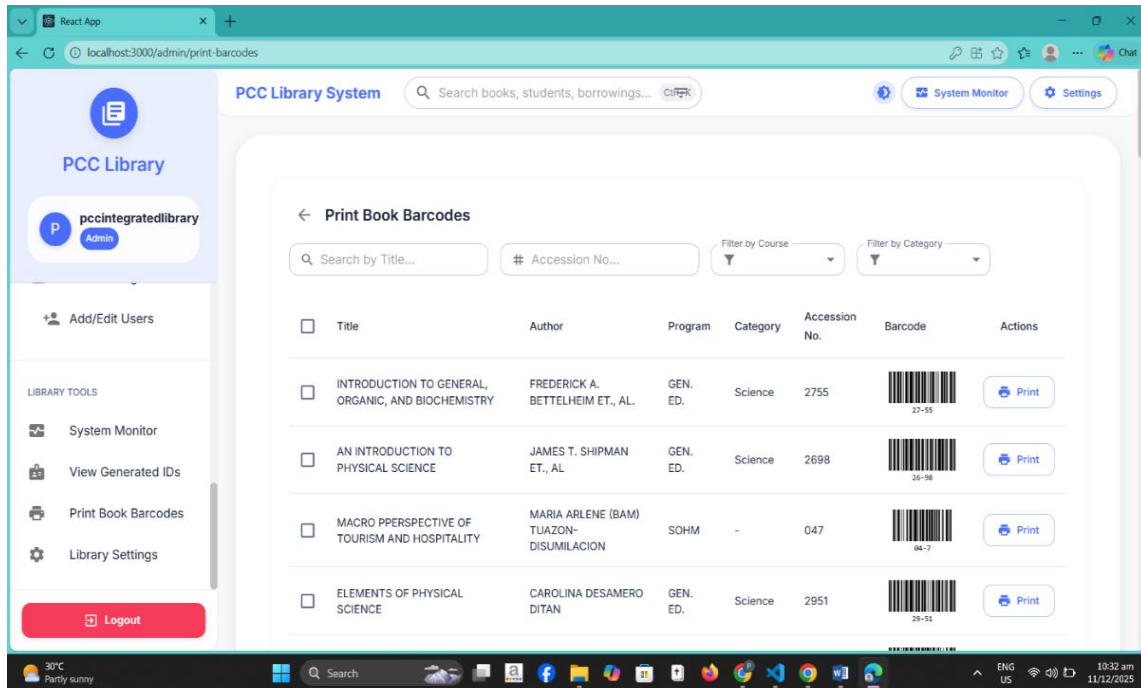


Figure 19. Design for Print Book Barcodes

Design for Library Settings.

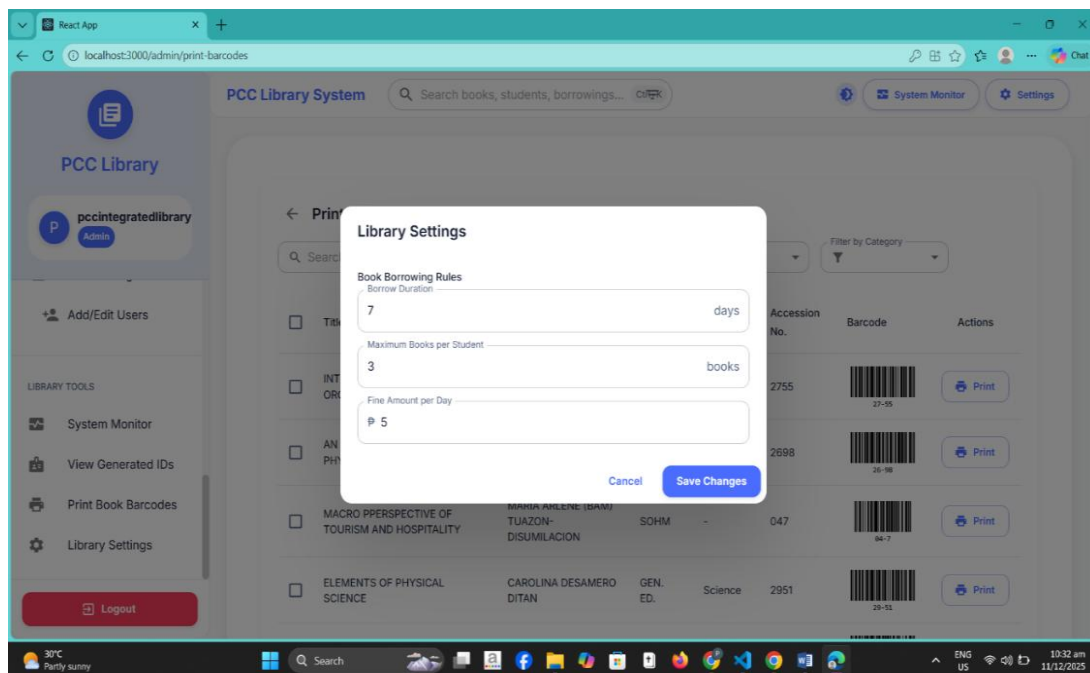
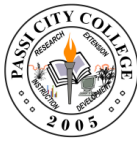


Figure 20. Design for Library Settings



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Coding. In this phase, the team developed the software according to the design specifications. This involved coding, testing, and debugging the software to ensure it meets the functional and technical requirements.

Code for User Interface.

```
1 import React, { useState, useEffect, useRef } from 'react';
2 import { useNavigate } from 'react-router-dom';
3 import {
4   Box,
5   TextField,
6   InputAdornment,
7   IconButton,
8   Paper,
9   Typography,
10  List,
11  ListItem,
12  ListItemText,
13  ListItemAvatar,
14  Avatar,
15  Popper,
16  ClickAwayListener,
17  Divider,
18  CircularProgress,
19  Chip,
20  useTheme,
21  alpha,
22  Button,
23  Menu,
24  MenuItem,
25  Tooltip,
26  ListItemButton,
27 } from '@mui/material';
28 import {
29   Search as SearchIcon,
30   Close as CloseIcon,
31   Book as BookIcon,
32   Person as PersonIcon,
33   SwapHoriz as BorrowIcon,
```

Figure 21. Code for User Interface



Code for Login Page.

```
1 import React, { useState, useEffect } from 'react';
2 import { useNavigate, useLocation } from 'react-router-dom';
3 import {
4   Container,
5   Paper,
6   Box,
7   TextField,
8   Button,
9   Typography,
10  Alert,
11  CircularProgress,
12  InputAdornment,
13  IconButton,
14  Link,
15  Dialog,
16  DialogTitle,
17  DialogContent,
18  DialogContentText,
19  DialogActions,
20 } from '@mui/material';
21 import {
22  Visibility,
23  VisibilityOff,
24  LibraryBooks,
25 } from '@mui/icons-material';
26 import { signInWithEmailAndPassword, sendPasswordResetEmail } from 'firebase/auth';
27 import { auth } from '.../../firebase';
28 import { useAuth } from '.../../contexts/AuthContext';
29 import { useThemeContext } from '.../../contexts/ThemeContext';
30
31 const Login: React.FC = () => {
32   const [email, setEmail] = useState('');
33   const [password, setPassword] = useState('');
```

Figure 22. Code for Login Page

Code for Dashboard.

```
1 import React, { useState, useEffect, useCallbck } from 'react';
2 import { useNavigate } from 'react-router-dom';
3 import {
4   Box,
5   Grid,
6   Paper,
7   Card,
8   CardContent,
9   Typography,
10  Avatar,
11  useTheme,
12  Button,
13  alpha,
14  IconButton,
15  Divider,
16  LinearProgress,
17  Tooltip, // Note: This import is unused in the new code, but kept for consistency
18  Stack, // Note: This import is unused in the new code, but kept for consistency
19 } from '@mui/material';
20
21 import {
22  LibraryBooks,
23  Group,
24  BookOnline,
25  Warning,
26  Book as BookIcon,
27  TrendingUp,
28  ArrowForward,
29  MenuBook, // Note: This import is unused in the new code, but kept for consistency
30  CalendarToday, // Note: This import is unused in the new code, but kept for consistency
31  MoreHoriz, // Note: This import is unused in the new code, but kept for consistency
32  PersonAdd, // Note: This import is unused in the new code, but kept for consistency
33 } from '@mui/icons-material';
34
35 const Dashboard: React.FC = () => {
36   // ...
37 }
```

Figure 23. Code for Dashboard



Code for Student Management.

```
68 const StudentManagement: React.FC = () => {
775   </Typography>
776   <Box sx={{ mt: 1, width: '100%', display: 'flex', justifyContent: 'center' }}>
777     <Barcode
778       value={selectedStudent.studentId}
779       width={1.5}
780       height={50}
781       fontSize={10}
782       margin={5}
783       background="#fff"
784     />
785   </Box>
786   <Typography variant="caption" display="block" sx={{ mt: 1, textAlign: 'center' }}>
787     {(selectedStudent.course) | (selectedStudent.yearLevel) (selectedStudent.section) ? ` ${selectedStudent.section} ` : ''}
788   </Typography>
789   </Card>
790 </form>
791 )}
792 </form>
793 </Box>
794 </DialogContent>
795 <DialogActions>
796   <Button onClick={() => {
797     setViewOpen(false);
798     setSelectedStudent(null);
799   }}>Close</Button>
800   <Button onClick={handlePrint} variant="contained" startIcon={PrintIcon}>Print ID
801   </Button>
802 </DialogActions>
803 </Dialog>
804 </Dialog>
805 <Dialog open={printBarcodesOpen} onClose={handleClosePrintBarcodes} maxWidth="md" fullWidth>
```

Figure 24. Code for Student Management

Code for Book Management.

```
1 import React, { useState, useEffect, useReducer, useImmer } from 'react';
2 import { useNavigate } from 'react-router-dom';
3 import {
4   Container,
5   Paper,
6   Table,
7   TableBody,
8   TableCell,
9   TableContainer,
10  TableHead,
11  TableFooter,
12  Button,
13  Dialog,
14  DialogTitle,
15  DialogContent,
16  DialogActions,
17  TextField,
18  IconButton,
19  Typography,
20  Box,
21  Grid,
22  Avatar,
23  CardMedia,
24  FormControl,
25  InputLabel,
26  Select,
27  MenuItem,
28  Chip,
29  Stack,
30  Autocomplete,
31  Checkbox,
32  Tooltip,
33  TablePagination,
```

Figure 25. Code for Book Management



Code for Borrow/Return.

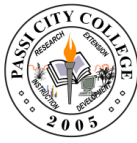
```
1 import React, { useState, useEffect, useCallback, useRef } from 'react';
2 import {
3   Box,
4   Paper,
5   Typography,
6   Button,
7   Grid,
8   Card,
9   CardContent,
10  Alert,
11  CircularProgress,
12  TextField,
13  InputAdornment,
14  List,
15  ListItem,
16  ListItemText,
17  IconButton,
18  Select,
19  MenuItem,
20  FormControl,
21  InputLabel,
22  Container,
23  Tabs,
24  Tab,
25  CardActions,
26  Divider,
27  Stack,
28  Dialog,
29  DialogTitle,
30  DialogContent,
31  DialogActions,
32  Input,
33  FormHelperText,
```

Figure 26. Code for Borrow/Return

Code for Borrowed Books.

```
1 import React, { useState, useEffect, useCallback } from 'react';
2 import {
3   Container,
4   Paper,
5   Typography,
6   Box,
7   TextField,
8   InputAdornment,
9   Table,
10  TableBody,
11  TableCell,
12  TableContainer,
13  TableHead,
14  TableHeader,
15  TableFooter,
16  Alert,
17  Chip,
18  Button,
19  IconButton,
20  Dialog,
21  DialogTitle,
22  DialogContent,
23  DialogActions,
24  Tooltip,
25 } from '@mui/material';
26 import {
27   Search as SearchIcon,
28   ArrowBack as ArrowBackIcon,
29   Remove as RemoveIcon,
30   Delete as DeleteIcon,
31 } from '@mui/icons-material';
32 import { ref, get, update, remove } from 'firebase/database';
33 import { database } from '../firebase';
```

Figure 27. Code for Borrowed Books



Code for Student Borrowing.

```
1 import React, { useState, useEffect, useCallba...
2 import {
3   Container,
4   Paper,
5   Typography,
6   Box,
7   TextField,
8   InputAdornment,
9   Accordion,
10  AccordionSummary,
11  AccordionDetails,
12  List,
13  ListItem,
14  ListItemText,
15  ListItemSecondaryAction,
16  CircularProgress,
17  Alert,
18  Chip,
19  Button,
20  IconButton,
21  Divider,
22  Avatar,
23  Tooltip,
24  Snackbar,
25  Dialog,
26  DialogTitle,
27  DialogContent,
28  DialogActions,
29  DialogContentText,
30 } from '@mui/material';
31 import {
32   Search as SearchIcon,
33   ArrowBack as ArrowBackIcon,
```

Figure 28. Code for Student Borrowing

Code for Reports.

```
76 const Reports: React.FC = () => {
914   <Button
915     onClick={exportToExcel}
916     startIcon={<ExcelIcon />}
917     variant="outlined"
918     color="success"
919     sx={{ mr: 1 }}
920   />
921   Export Excel
922   </Button>
923   <Button
924     onClick={exportToPDF}
925     startIcon={<PrintIcon />}
926     variant="contained"
927     color="primary"
928   />
929   Generate PDF
930   </Button>
931   </>
932   </DialogActions>
933   </Dialog>
934   </Paper>
935   </Container>
936   </>
937   </>
938   </>
939   </>
940   export default Reports;
```

Figure 29. Code for Reports



Code for Announcements.

```
1 import React, { useState, useEffect } from 'react';
2 import {
3   Container,
4   Paper,
5   Typography,
6   TextField,
7   Button,
8   FormControl,
9   InputLabel,
10  Select,
11  MenuItem,
12  Box,
13  Grid,
14  Alert,
15  Chip,
16  CircularProgress,
17  Table,
18  TableBody,
19  TableCell,
20  TableContainer,
21  TableHead,
22  TableFooter,
23  Checkbox,
24  Card,
25  CardContent,
26  CardMedia
27 } from '@mui/material';
28 import { ref, get } from 'firebase/database';
29 import { database } from '../firebase';
30 import { sendAnnouncement } from '../utils/notificationUtils';
31
32 interface Student {
33   id: string;
34   name: string;
35 }
```

Figure 30. Code for Announcements

Code for Overdue Notifications.

```
1 import React, { useState, useEffect } from 'react';
2 import {
3   Box,
4   Button,
5   Card,
6   CardContent,
7   Dialog,
8   DialogTitle,
9   Divider,
10  FormControlLabel,
11  Grid,
12  IconButton,
13  Paper,
14  Table,
15  TableBody,
16  TableCell,
17  TableContainer,
18  TableHead,
19  TableFooter,
20  Typography,
21  CircularProgress,
22  Alert,
23  Snackbar,
24  EmailIcon,
25  CloseIcon,
26  RefreshIcon,
27  SendIcon
28 } from '@mui/material';
29 import { ref, get } from 'firebase/database';
30 import { database } from '../firebase';
31 import { sendOverdueNotification } from '../utils/notificationUtils';
32
33 interface Student {
34   id: string;
35   name: string;
36 }
```

Figure 31. Code for Overdue Notifications



Code for Fine Payments.

```
1 import React, { useState, useeffect, useCallack } from 'react';
2 import {
3   Box,
4   Paper,
5   Typography,
6   Button,
7   TextField,
8   Grid,
9   Card,
10  CardContent,
11  IconButton,
12  Table,
13  TableBody,
14  TableCell,
15  TableContainer,
16  TableHead,
17  TableRow,
18  Divider,
19  Dialog,
20  DialogTitle,
21  DialogContent,
22  DialogActions,
23  Alert,
24  CircularProgress,
25  Chip,
26  FormControl,
27  InputLabel,
28  Select,
29  MenuItem,
30  Tooltip,
31  InputAdornment,
32 } from '@mui/material';
33 import {
```

Figure 32. Code for Fine Payments

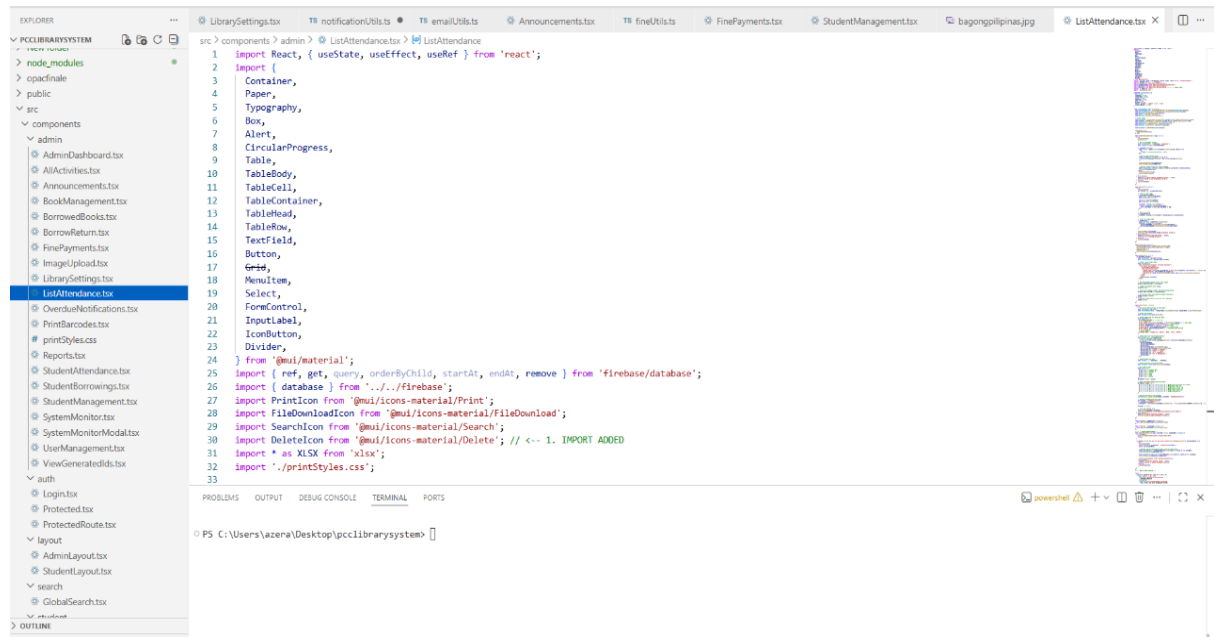
Code for Record Attendance.

```
1 import React, { useState, useeffect } from 'react';
2 import {
3   Container,
4   Paper,
5   Typography,
6   Box,
7   Alert,
8   CircularProgress,
9   List,
10  ListItem,
11  ListItemText,
12  Divider,
13  Card,
14  CardContent,
15  Grid,
16  IconButton,
17  TextField,
18  Button,
19  Switch,
20  FormControlLabel,
21 } from '@mui/material';
22 import DeleteIcon from '@mui/icons-material/Delete';
23 import { ref, get, push, query, orderByChild, equalTo, remove, set } from 'firebase/database';
24 import { database } from '../firebase';
25
26 interface Student {
27   id: string;
28   name: string;
29   studentId: string;
30   course: string;
31   barcode: string;
32 }
33
```

Figure 33. Code for Record Attendance



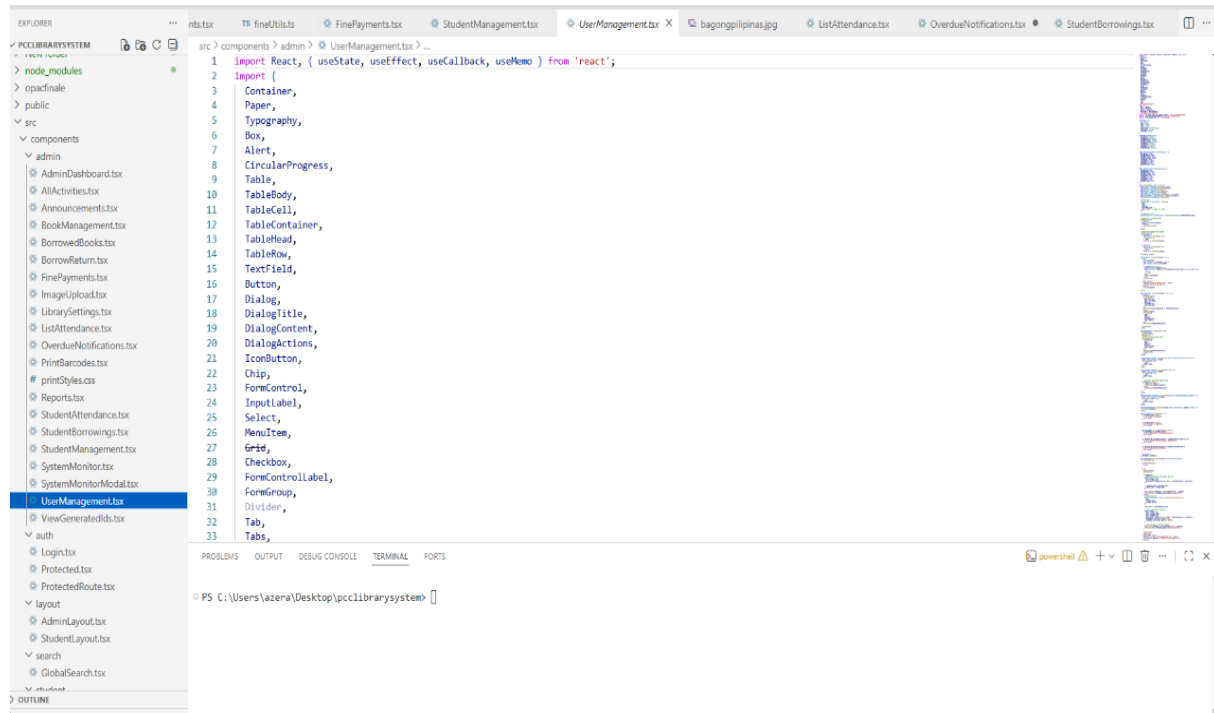
Code for List and Print Attendance.



```
1 import React, { useState, useEffect, useRef } from 'react';
2 import {
3   Container,
4   Paper,
5   Typography,
6   Box,
7   Alert,
8   CircularProgress,
9   Table,
10  TableBody,
11  TableCell,
12  TableContainer,
13  TableHead,
14  TableRow,
15  TextField,
16  Button,
17  Grid,
18  MenuItem,
19  Select,
20  FormControl,
21  InputLabel,
22  IconButton,
23  Divider,
24 } from '@mui/material';
25 import { ref, get, query, orderByChild, startAt, endAt, remove } from 'firebase/database';
26 import { database } from '../firebase';
27 import PrintIcon from '@mui/icons-material/Print';
28 import FileDownloadIcon from '@mui/icons-material/FileDownload';
29 import SearchIcon from '@mui/icons-material/Search';
30 import DeleteIcon from '@mui/icons-material/Delete'; // <-- 1. IMPORT ADDED
31 import * as XLSX from 'xlsx';
32 import './printStyles.css';
33
```

Figure 34. Code for List & Print Attendance

Code for User Management.



```
1 import React, { useState, useEffect, useCallback, useMemo } from 'react';
2 import {
3   Container,
4   Paper,
5   Typography,
6   Box,
7   Alert,
8   CircularProgress,
9   Table,
10  TableBody,
11  TableCell,
12  TableContainer,
13  TableHead,
14  TableRow,
15  TextField,
16  Button,
17  Dialog,
18  DialogTitle,
19  DialogContent,
20  DialogActions,
21  IconButton,
22  Chip,
23  FormControl,
24  InputLabel,
25  Select,
26  MenuItem,
27  Grid,
28  Checkbox,
29  FormControlLabel,
30  FormGroup,
31  Divider,
32  Tab,
33  Tabs,
34
```

Figure 35. Code for User Management



Code for System Monitor.

```
1 import React, { useState, useEffect, useCallba... } from 'react';
2 import {
3   Box,
4   Paper,
5   Typography,
6   Grid,
7   CircularProgress,
8   Card,
9   CardContent,
10  Button,
11  Divider,
12  TextField,
13  MenuItem,
14  Alert,
15  Tooltip as MuiTooltip,
16  Avatar,
17  Stack,
18  Chip,
19 } from '@mui/material';
20 import {
21   NetworkCheck as NetworkIcon,
22   Speed as SpeedIcon,
23   Memory as MemoryIcon,
24   Storage as StorageIcon,
25   Sync as SyncIcon,
26   People as PeopleIcon,
27   Person as PersonIcon,
28   Group as GroupIcon,
29   CheckCircle as CheckCircleIcon,
30   Cancel as CancelIcon,
31 } from '@mui/icons-material';
32 import { LineChart, Line, XAxis, YAxis, CartesianGrid, Tooltip as RechartsTooltip, Legend, ResponsiveContainer } from 'recharts';
33 import { database } from '../firebase';
34
35 ...
```

Figure 36. Code for System Monitor

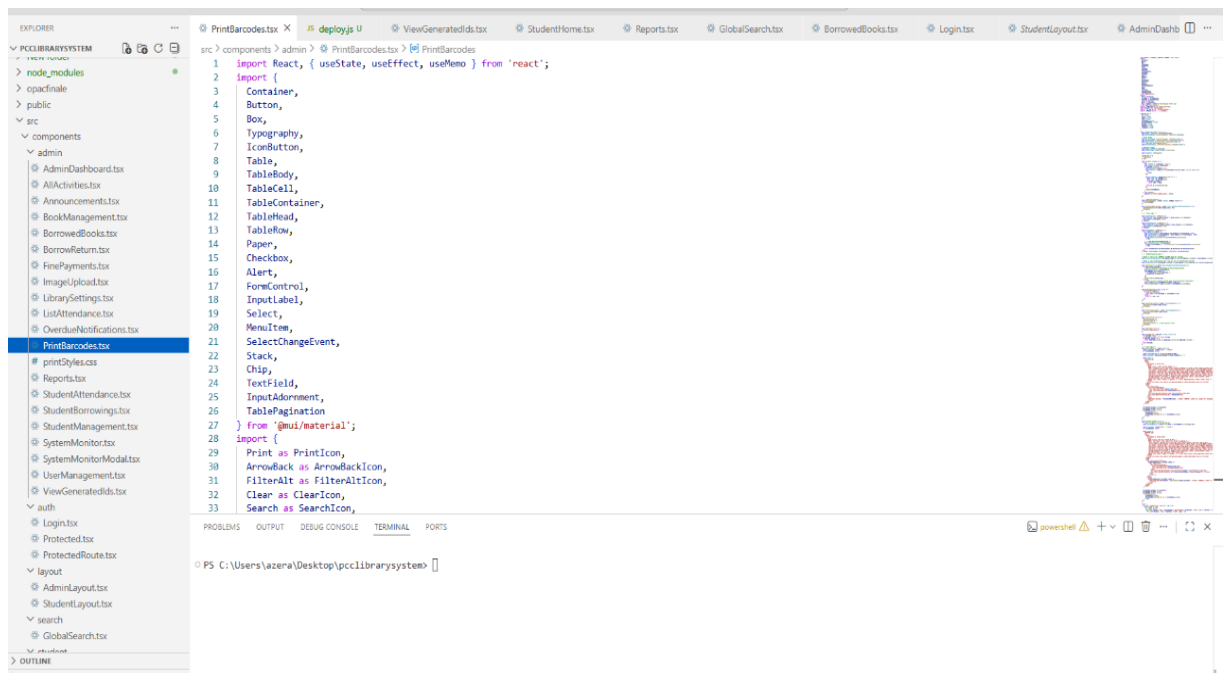
Code for View Generated IDs.

```
1 import React, { useState, useEffect, useCallba... } from 'react';
2 import {
3   Dialog,
4   DialogTitle,
5   DialogContent,
6   DialogActions,
7   Button,
8   Box,
9   Card,
10  Typography,
11  IconButton,
12  Table,
13  TableBody,
14  TableCell,
15  TableContainer,
16  TableHead,
17  TableFooter,
18  Paper,
19  Checkbox,
20  TextField,
21  InputAdornment,
22  MenuItem,
23  Select,
24  FormControl,
25  InputLabel,
26  SelectChangeEvent, // Added for type safety
27 } from '@mui/material';
28 import { Print as PrintIcon, Search as SearchIcon, FilterList as FilterIcon } from '@mui/icons-material';
29 import Barcode from 'react-barcode';
30 import { ref, get } from 'firebase/database';
31 import { database } from '../firebase';
32 import './printStyles.css';
33
34 ...
```

Figure 37. Code for View Generated IDs



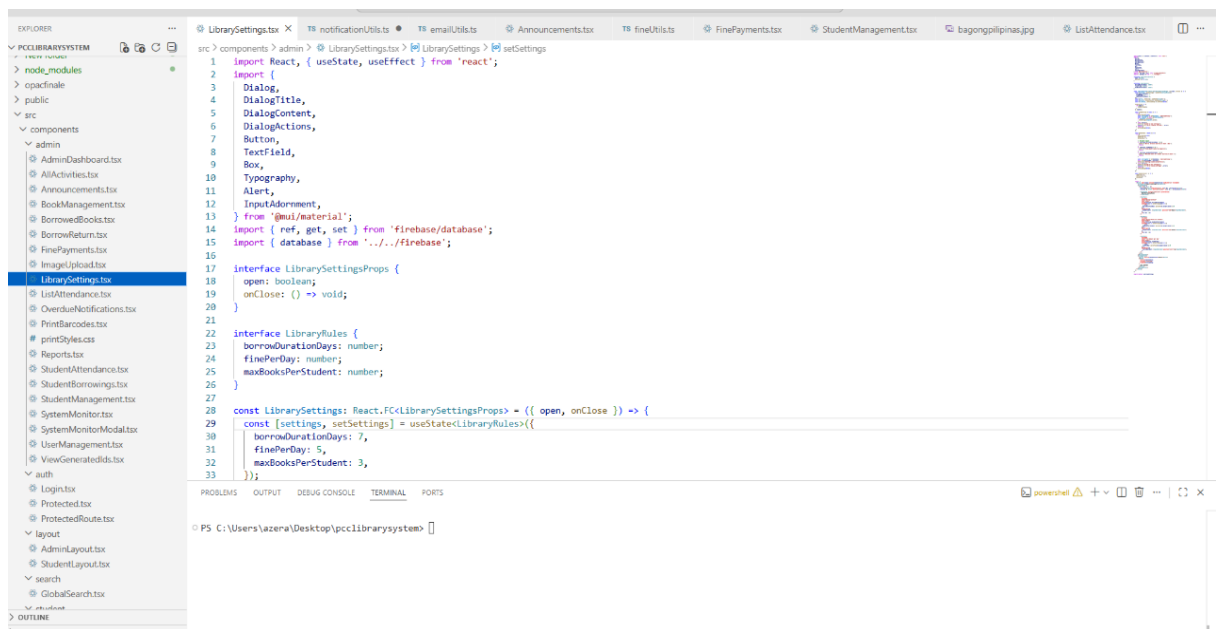
Code for Print Barcodes.



```
1 import React, { useState, useEffect, useMemo } from 'react';
2 import {
3   Container,
4   Button,
5   Box,
6   Typography,
7   IconButton,
8   Table,
9   TableBody,
10  TableCell,
11  TableContainer,
12  TableHead,
13  TableRow,
14  Paper,
15  Checkbox,
16  Alert,
17  FormControl,
18  InputLabel,
19  Select,
20  MenuItem,
21  SelectChangeEvent,
22  Stack,
23  Chip,
24  TextField,
25  InputAdornment,
26  TablePagination
27 } from '@mui/material';
28 import {
29   Print as PrintIcon,
30   ArrowBack as ArrowBackIcon,
31   FilterAlt as FilterAltIcon,
32   Clear as ClearIcon,
33   Search as SearchIcon,
```

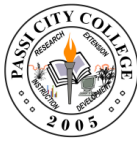
Figure 38. Code for Print Barcodes

Code for Library Settings.



```
1 import React, { useState, useEffect } from 'react';
2 import {
3   Dialog,
4   DialogTitle,
5   DialogContent,
6   DialogActions,
7   Button,
8   TextField,
9   Box,
10  Typography,
11  Alert,
12  InputAdornment,
13 } from '@mui/material';
14 import { ref, get, set } from 'firebase/database';
15 import { database } from '../firebase';
16
17 interface LibrarySettingsProps {
18   open: boolean;
19   onClose: () => void;
20 }
21
22 interface LibraryRules {
23   borrowDurationDays: number;
24   finePerDay: number;
25   maxBooksPerStudent: number;
26 }
27
28 const LibrarySettings: React.FC<LibrarySettingsProps> = ({ open, onClose }) => {
29   const [settings, setSettings] = useState<LibraryRules>({
30     borrowDurationDays: 7,
31     finePerDay: 5,
32     maxBooksPerStudent: 3,
33   });
```

Figure 39. Code for Library Settings



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The following specific tasks were implemented:

1. Database Development

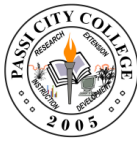
- Designed and implemented a relational database to store book records, user accounts, borrowing history, and search logs.
- Created tables for books, students, user, and transactions.
- Developed SQL queries for efficient data retrieval, insertion, updating, and deletion.

2. OPAC Development

- Developed an interactive OPAC allowing users to search for books using filters such as title, author, subject, and category.
- Implemented an auto-suggestion feature to enhance search accuracy.
- Enabled real-time availability checking for books in the library.

3. User Authentication & Role Management

- Developed a secure login system for administrators.



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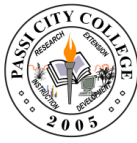
- Implemented role-based access control (RBAC) to restrict access to specific features based on admin roles.
- Integrated two-factor authentication (2FA) for enhanced security (if applicable).

4. Book Borrowing & Return System

- Developed a module to process book borrowing and returning transactions.
- Implemented an automated due date calculator and penalty system for overdue books.
- Sent email or SMS notifications for due date reminders and overdue penalties.

5. Library Catalog Management

- Created a dashboard for librarians to manage book records (adding, editing, and deleting books).
- Integrated barcode code scanning for quick book lookup and check-out.
- Implemented a bulk book import/export feature for easy catalog updates.



6. Reporting & Analytics

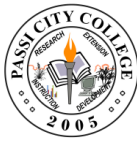
- Developed a reporting module to generate statistics on book borrowing trends, most searched books, and overdue items.
- Integrated graphical reports and charts for better visualization of library usage.
- Enabled exporting reports to PDF/Excel for administrative use.

7. System Testing & Debugging

- Conducted unit testing to verify the functionality of individual modules.
- Performed integration testing to ensure smooth interaction between different system components.
- Conducted user acceptance testing (UAT) with students and librarians to validate usability and performance.

8. Deployment & Maintenance

- Deployed the system on local servers/cloud hosting for accessibility within Passi City College.
- Provided training and documentation for librarians and users on how to use the OPAC.



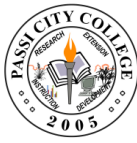
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- o Set up a maintenance plan to address future updates and technical support.

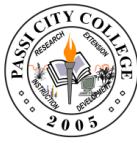
Testing. The code is then handed over to the testing team. Testers check the program for all possible defects by running test cases either manually or through automation. The client is involved in the testing phase as well to ensure all requirements are met. All flaws and bugs detected during this phase are fixed to ensure Quality Assurance.

Deployment. In this phase, the system was deployed on a local host environment at Passi City College to ensure it was fully functional and ready for real-world use. The deployment involved setting up a local server, configuring the necessary software dependencies, and ensuring that the system's components worked seamlessly together. After completing the initial setup, rigorous testing was carried out to verify that all functionalities were operating as expected, including performance checks to ensure the system could handle expected loads.



Maintenance. Once the Smart Library System with Online Public Access Catalog (OPAC) was deployed, the team provided ongoing support and maintenance to ensure the system remains functional and up-to-date. This includes fixing bugs, adding new features, and providing user support.

Overall, the SDLC provided a structured approach to developing the Smart Library System with Online Public Access Catalog for Passi City College, ensuring that the system meets the requirements and is delivered on time and within budget.



Project Requirements Specification

i. Operational Feasibility

Functional Decomposition Diagram

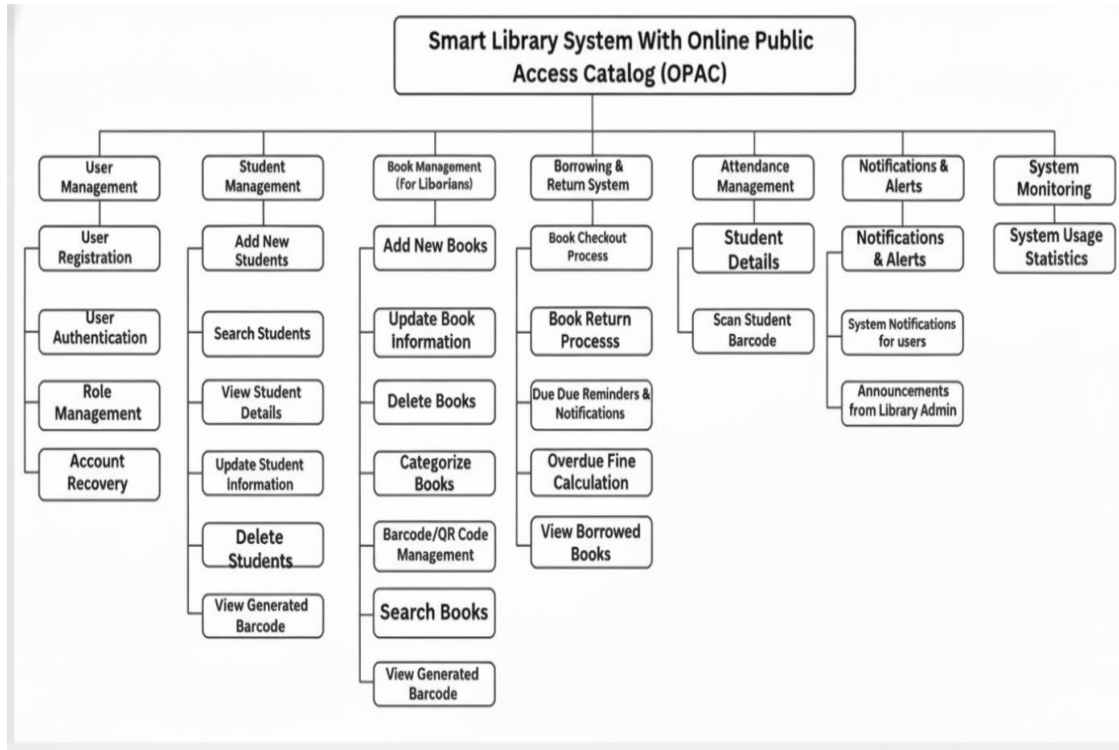
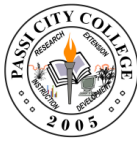


Figure 40. Operational Feasibility Functional Decomposition Diagram

The functional decomposition diagram provides a detailed breakdown of the Smart Library System, illustrating how the systems complex operations are organized into smaller, manageable components. This diagram captures the flow and interactions of task from the high-level system functions down to specific, actionable task. It demonstrates the distinct roles of Administrator (librarian) and the User (students), highlighting how the system automates the



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borrowing and returning processes while maintaining accurate inventory records. Show in Figure 40.

Use Case Diagram

The study used a Use Case Diagram in order to describe how the Student and the Librarian interact with the system. It shows that the student needs to visit the library website URL to search for books via the Online Public Access Catalog (OPAC), and view the books availability. For the Librarian, the process is different from what the student does, because only the Librarian can register new students, generate students library Ids, add new books to the inventory, process borrow and return transactions, can send notifications of the students transaction, can record attendance, generate printed reports, and access the system monitoring dashboard for security purposes.

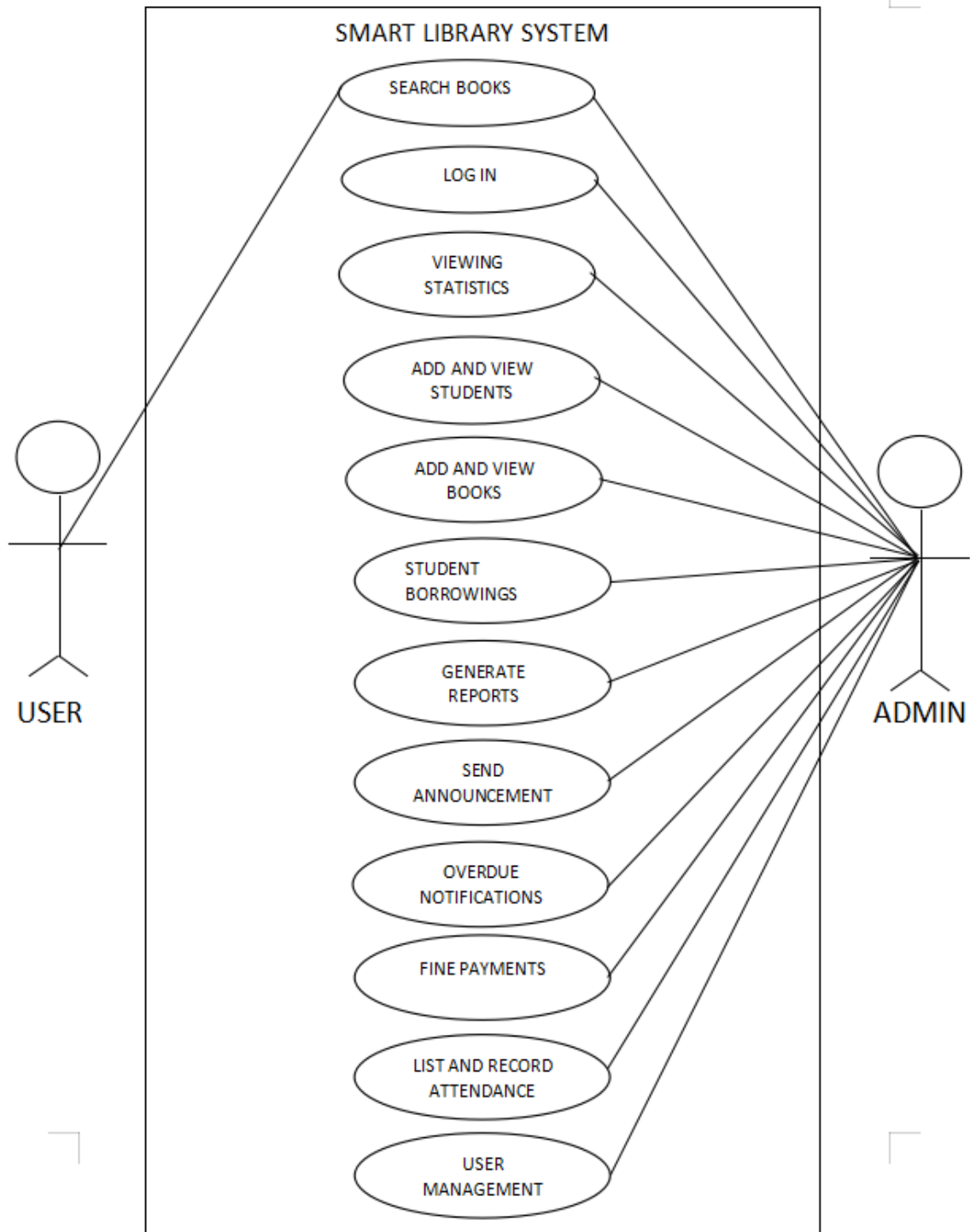


Figure 41. Use Case Diagram



ii. Technical Feasibility

Software Specification

The proposed Smart Library System integrated with an Online Public Access Catalog (OPAC) provides an enhanced level of user convenience and efficient transactions. The system ensures seamless management of library resources, making it a practical and viable solution for Passi City College. The development team has determined that the project is technically and operationally feasible. This system is built with the following specifications:

- Operating System: Windows 10 Pro
- Database Management System: Firebase

Hardware Specification

Processor: Acer Aspire 5 Intel Core i5

Installed RAM: 8GB DDR4

System Type: 64-bit x64 operating system

Storage: 256GB SSD (Solid State Drive)

Hardware Used: Handheld Barcode Scanner

Interface: USB 2.0 (Plug and Play / HID)

Scan Technology: Laser or CCD (Charge-Coupled Device)

Scanning Speed: Minimum 100 scans per second



iii. Schedule Feasibility

Gantt Chart

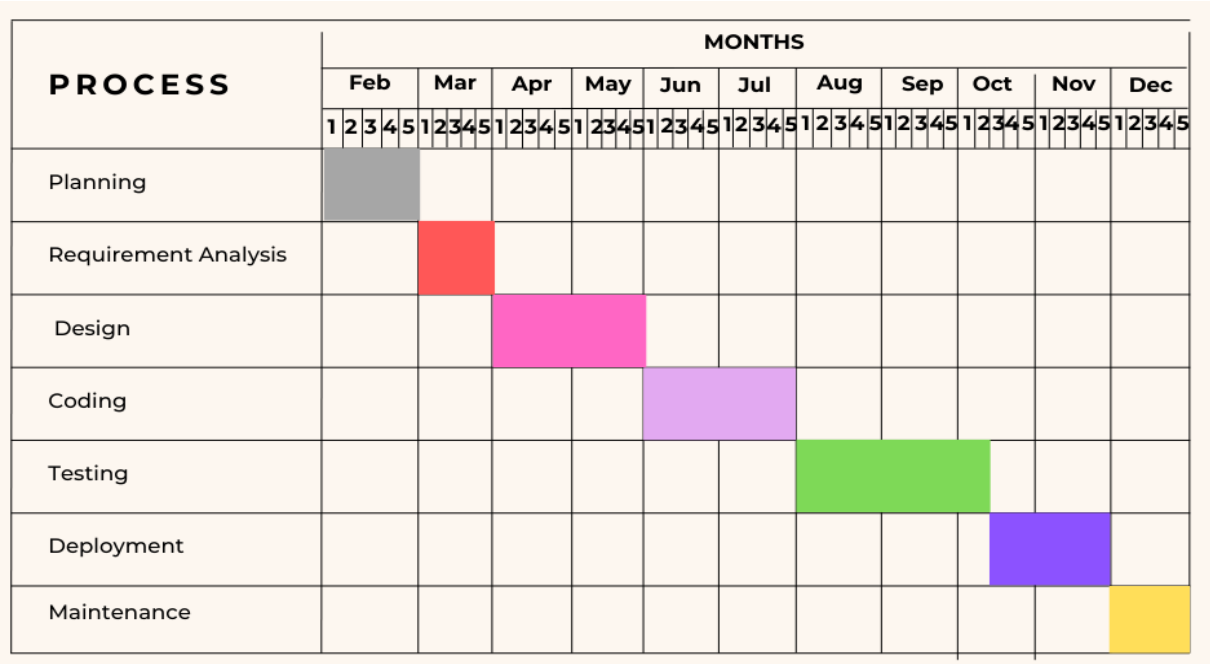


Table 1. Gantt Chart of Smart Library System

The researcher utilized the Gantt Chart as a planning and tracking tool to ensure the timely development of the Smart Library System integrated with an Online Public Access Catalog (OPAC). This chart outlined various key activities (as shown in Table 2), helping to evaluate the feasibility and attainability of the project within the given time constraints. It also facilitated assessing resource availability, identifying dependencies, and anticipating potential risks, ensuring that the system could be successfully completed within the planned schedule.

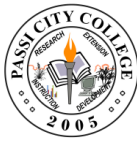


iv. Economic feasibility

Cost and Benefit Analysis

Due to the absence of a computerized library management system at Passi City College, the research team aimed to develop a web-based application. This system was designed to streamline library operations, minimizing the reliance on manual record-keeping and enhancing the efficiency of book inventory management and transactions. The traditional manual process of managing library resources was both time-consuming and costly. In contrast, the estimated cost of the proposed

Smart Library System integrated with an Online Public Access Catalog (OPAC) was significantly lower. Once fully implemented, the system would allow the administration to operate efficiently without incurring substantial costs in the subsequent years.



Development Cost of the Project

Project Task	Cost
System Development	₱ 12,000.00
Hardware Used (Barcode Scanner)	₱ 500.00
Miscellaneous	₱ 5,000.00
Total:	₱ 17,500.00

Table 2. Development Cost of the Project

Project Design

i. Requirements Modeling

IPO Chart

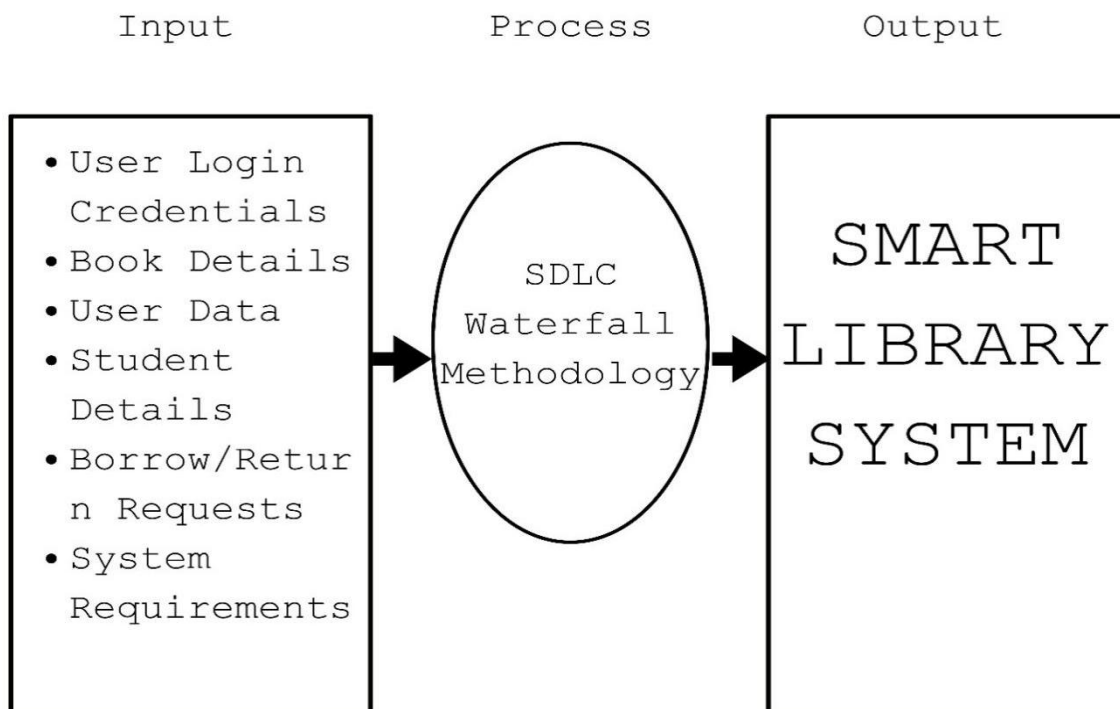


Figure 42. Input-Process-Output



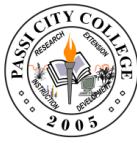
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The Input-Process-Output (IPO) Diagram was utilized in this study, as illustrated in Figure 41, to represent the study's variables. The Smart Library System integrated with an Online Public Access Catalog (OPAC) accepted book and user information as input, processed these details by organizing data, tracking book transactions, and generating reports, and produced outputs such as library usage statistics and inventory summaries.

This comprehensive system enhanced library resource management, ensured quick access to book records, and supported data-driven decision-making for both users and administrators at Passi City College.

The researchers followed the System Development Life Cycle (SDLC) methodology in designing and implementing the system. The application underwent usability testing to evaluate its functionality and effectiveness. The proposed Smart Library System with OPAC, along with its evaluation results, constituted the final output of the study.



Context Diagram

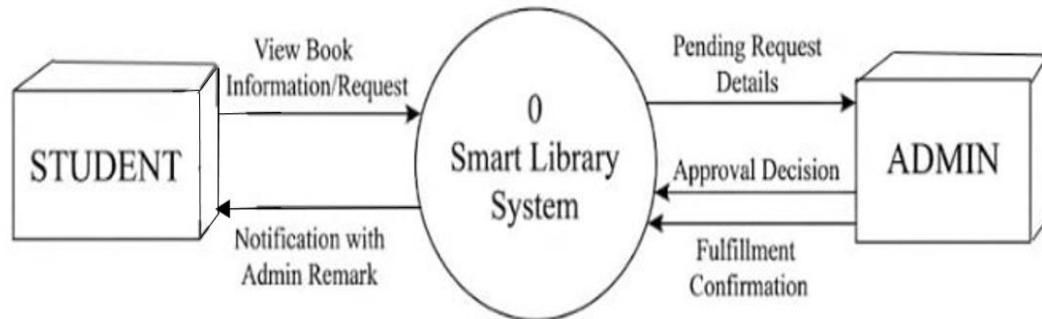


Figure 43. Context Diagram

The context diagram depicted the scope of the Smart Library System integrated with an Online Public Access Catalog (OPAC) and illustrated the external entities, including users and administrators, that interacted with it via the web-based platform. Users, such as students and faculty, could search for and view book details, availability, and other library resources. Meanwhile, administrators had additional privileges, allowing them to manage book records, update inventory, and oversee system operations. The system ensured seamless access to library materials and efficient resource management.

Data Flow Diagram

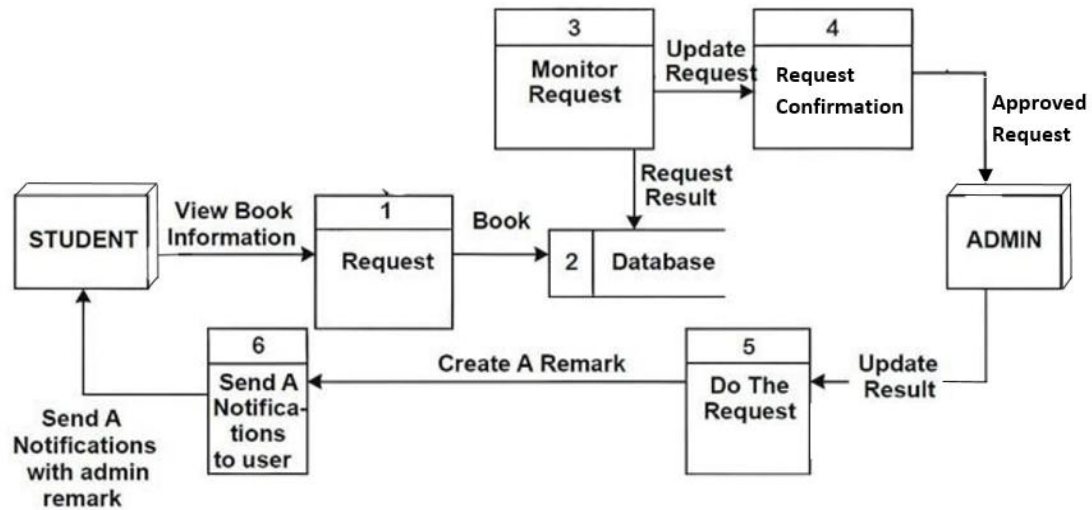


Figure 44. Data Flow Diagram

The diagram demonstrated the process where a user initially submits a request by specifying their library service needs, which are then recorded in the database. The administrator reviews and approves the request before proceeding with its fulfillment. Once completed, the administrator provides feedback, including relevant remarks, to the user. Additionally, the administrator has the capability to track and monitor the status of the request throughout the process.



System Flowchart

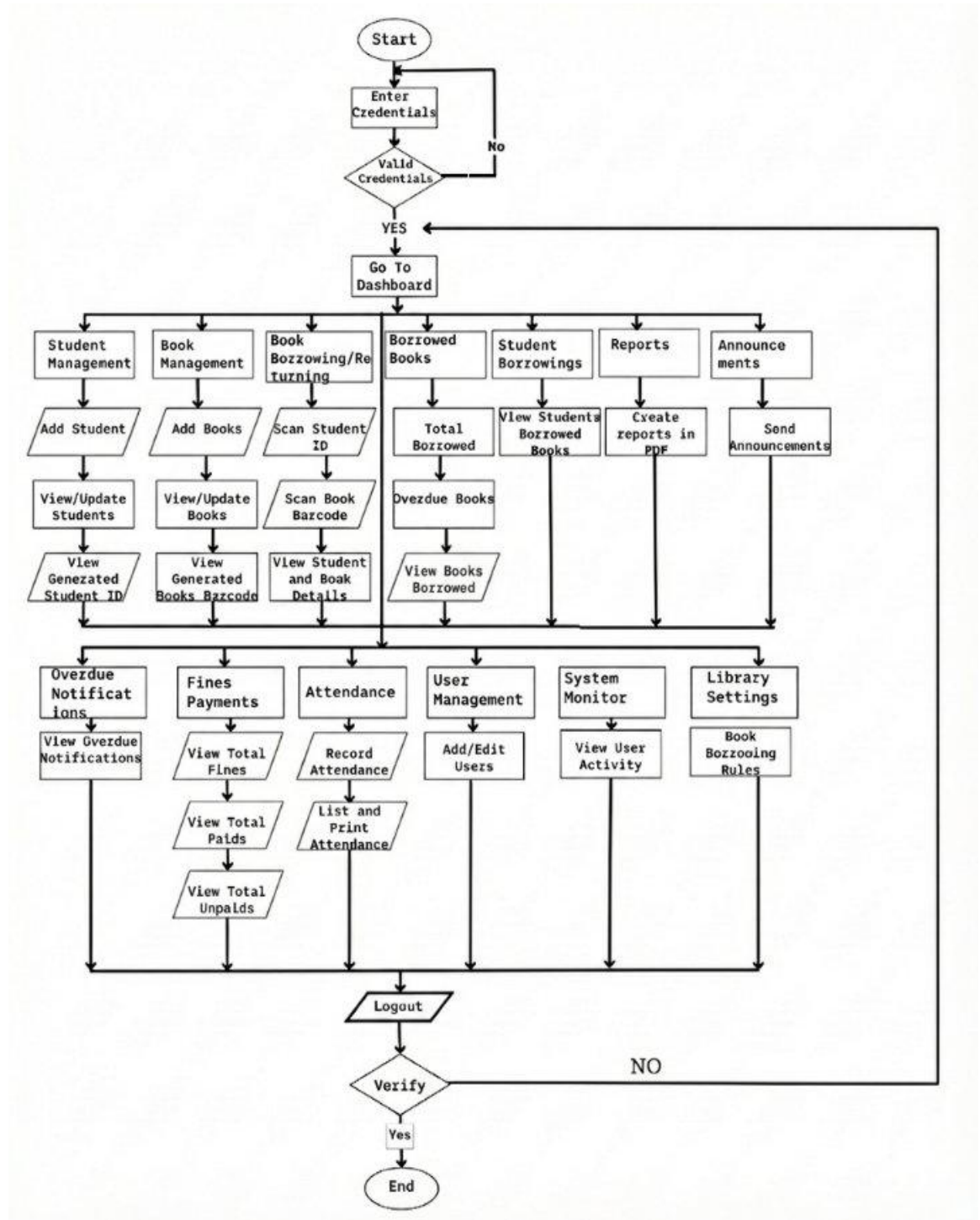


Figure 45. Admin System Flowchart



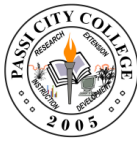
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The system flowchart outlined the operational structure of the Smart Library System with Online Public Access Catalog (OPAC), illustrating the workflow for administrators as they navigate its features. The process starts with an admin login, where credentials are verified before access is granted. Upon successful authentication, the administrator is directed to the main dashboard, which provides an overview of the system, including the total number of books, users, transactions, and reservations.

From the dashboard, the administrator can access various essential modules to manage different aspects of the library system. In the Books Management section, the admin can add new books or update existing records, ensuring that the catalog remains accurate and up to date. The User Management module allows the admin to register new users and manage their access. The Category and Classification section enables the addition, viewing, and modification of book categories and classifications, helping to streamline the library's organizational structure.

The Requests and Reservations section handles book reservations and borrowing requests, where the administrator can approve or decline requests based on availability. The



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Reports and Analytics module provide access to reports related to book lending, overdue items, and user activity, allowing for informed decision-making and efficient library management.

After completing administrative tasks, the admin can securely log out of the system. A confirmation prompt ensures secure session termination, preventing unauthorized access. This well-structured workflow enhances efficiency, improves user experience, and optimizes the management of Passi City College's Smart Library System.



ii. Data Design

Entity Relationship Diagram

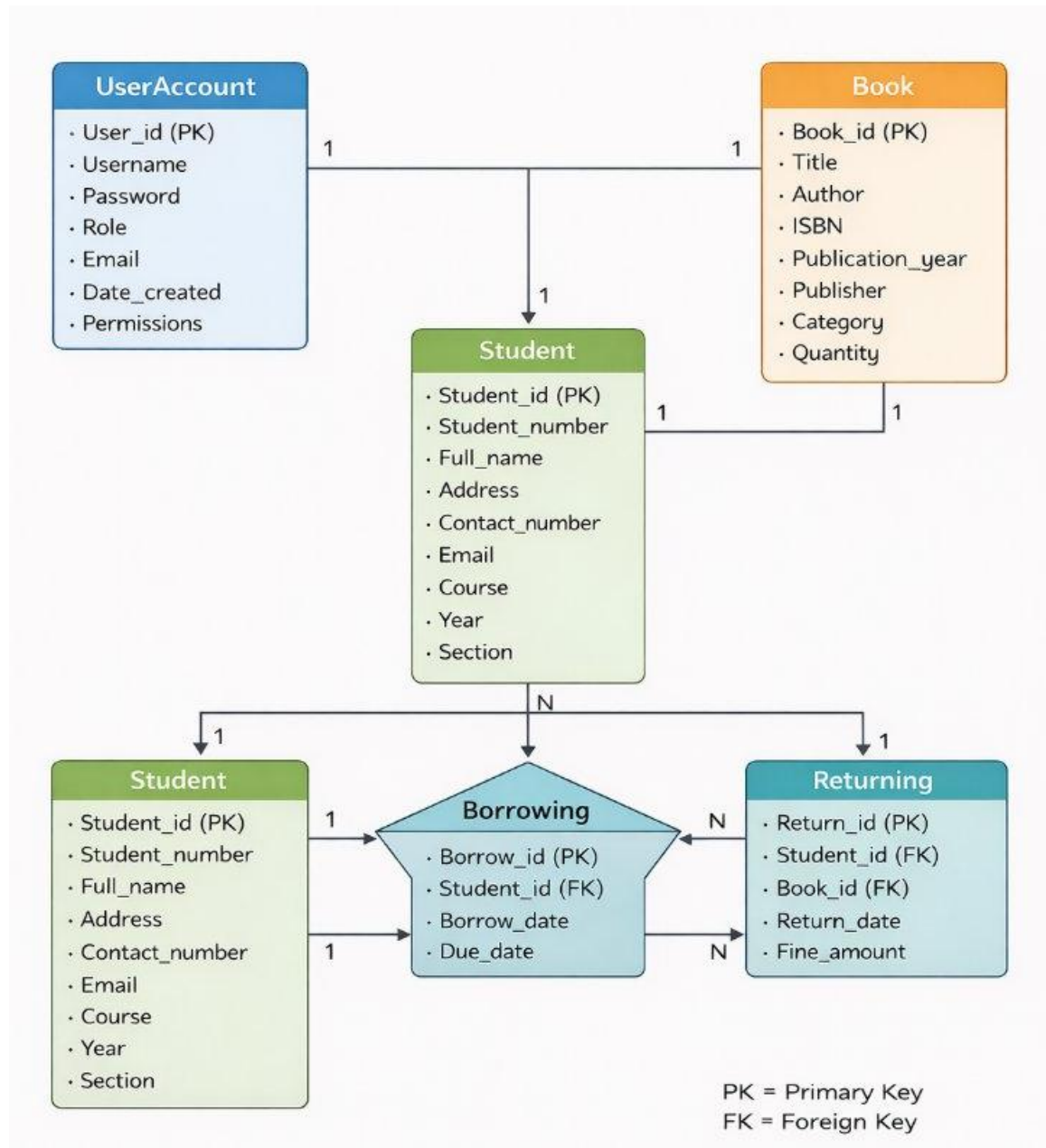


Figure 46. Entity Relationship Diagram



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The Entity Relationship Diagram (ERD) for the Smart Library System with Online Public Access Catalog (OPAC) represents various entities, each with distinct attributes to support efficient library management.

The Books entity stores essential details such as `book_id`, `title`, `author`, `category`, `ISBN`, `status`, and `location`. The Users entity contains `user_id`, `name`, `role`, `contact_info`, and `action`, representing students, faculty, and staff utilizing the system. The Categories entity includes `category_id`, `category_name`, and `description`, helping organize books efficiently.

The Transactions entity tracks borrowing and returning activities, storing attributes such as `transaction_id`, `user_id`, `book_id`, `borrow_date`, `return_date` and `status`. The Reports entity generates insights into book circulation, overdue items, and system activity, maintaining `report_id`, `report_type`, `generated_date` and `action`.

Relationships among these entities are established using foreign keys, ensuring seamless integration between book tracking, user interactions, and transaction history. The Transactions entities link users with book borrowing and returning processes, while the Reports entity provides



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analytical insights for monitoring library operations. The structured database design promotes data consistency, efficient record-keeping, and improved accessibility of library resources.

Data Dictionary

Users Table

Field Name	Description	Data Type	Size	Constraints
User_ID	Unique identifier for user	INT	11	PK. AUTO- INCREMENT
Full_ name	Full name of the user	VARCHAR	150	NOT NULL
Role	Role of the user (Student, Faculty, Librarian, Admin)	ENUM	-	UNIQUE
Email	User's email address	VARCHAR	100	NOT NULL
Password	Hashed password for login	VARCHAR	25	

Table 3. Users Data Dictionary



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Books Table

Field Name	Description	Data Type	Size
Book_ID	Hashed password for login	INT	11
Title	Title of the book	VARCHAR	150
Author	Author of the book	VARCHAR	100
ISBN	International Standard Book Number	VARCHAR	20
Category	Book category/genre	VARCHAR	50
Published_Year	Year of publication	YEAR	-
Status	Availability status (Available/Issued	ENUM	-

Table 4. Book Data Dictionary



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Student Table

Field Name	Description	Data Type	Size	Constraints
Student_ ID	Unique identifier for each student	INT	11	PK , AUTO_INCREMENT
User_ ID	Links student to general Users table	INT	11	11 FK Users (User_ID)
Student_ No	Unique student number (e.g., ID number)	VARCHAR	20	UNIQUE, NOT NULL
Course	Academic program/course enrolled	VARCHAR	10	NOT NULL
Year Level	Current year level (1st, 2nd, 3rd, 4th)	INT	1	NOT NULL
Section	Student's section or class group	VARCHAR	50	NOT NULL
Date_ Joined	Date student was registered in system	DATE	—	DEFAULT CURRENT_YEAR

Table 5. Student Data Dictionary



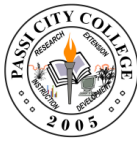
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Transaction Table

Field Name	Data Type	Key	Description
Transaction_ID	INT	PK	Unique identifier for each transaction
Users_ID	INT	PK	References the Users table (who performed the transaction)
Book_ID	INT NULL		References the Books table (if applicable, e.g., borrow/return)
Action_Type	VARCHAR(50)		Type of action (e.g., "Borrow", "Return", "Search", "Fine")
Action_Date	DATE/TIME		Date and time of the transaction
Status	VARCHAR (20)		Status of the transaction (e.g., "Completed", "pending")
Remarks	VARCHAR (255)		Additional notes (optional)

Table 6. Transaction Data Dictionary



iii. Output and User- interface Design

The image shows a login page for the OPAC System. At the top center is a blue icon of a book with a magnifying glass. Below it, the text "OPAC System" is displayed in a large, bold, black font. Underneath, a smaller line of text reads "Sign in to access the library management system". There are two input fields: the first is labeled "Email Address *" and contains a single vertical line; the second is labeled "Password *" and has a small eye icon to its right. Below these fields is a solid blue button with the text "Sign In" in white. At the bottom of the page, a line of text states "Only authorized users can access this system."

Figure 47. Login Page

Admin/Staff type their username and password to login the system.

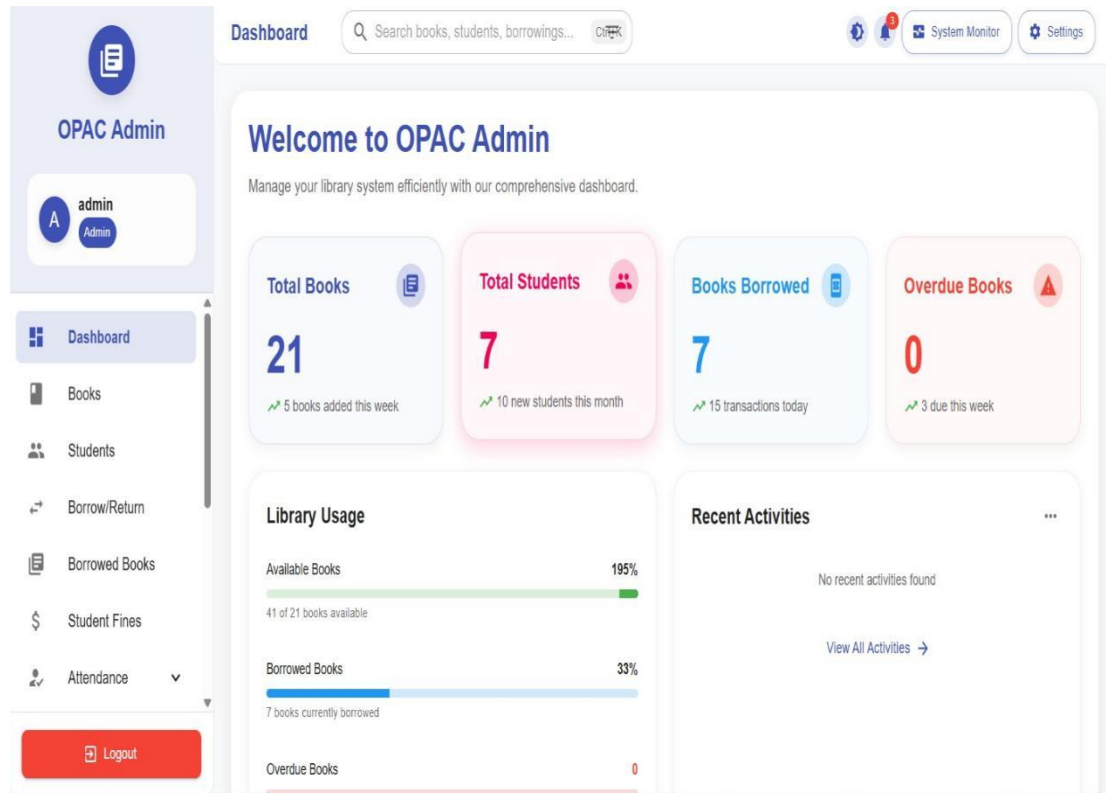


Figure 48. Admin Dashboard Page

Dashboard displays the brief description of the system, the total books, total students, books borrowed and overdue books.



The screenshot shows the 'Students' section of the OPAC Admin interface. On the left is a sidebar with navigation links: Dashboard, Students (selected), Books, Borrow/Return, and Borrowed Books. The main content area is titled 'Student Management' and includes a search bar, a 'Generated Student IDs' button, a 'Print Selected Barcodes' button (showing 0), and an 'Add New Student' button. Below these are filters for Course, Year Level, and Section, along with a 'Clear Filters' button. A table lists student records with columns for checkboxes, Student ID, Barcode, Name, Address, Email, and Course.

☐	Student ID	Barcode	Name	Address	Email	Course
☐	202573797318		kris	bugtongan	arcampsfers@gmail.com	Bachelor of Science in Information Technology
☐	202593591626		cherry	passi	aianoelarguelles94@gmail.com	Bachelor of Science in Information

Figure 49. Student Page

The screenshot shows the 'Books' section of the OPAC Admin interface. The sidebar is identical to the previous page. The main content area is titled 'Book Management' and includes a 'Print Barcode' button and an 'Add New Book' button. Below these are filter buttons for Author, Copyright Year, Publication, and Category. It indicates 'Showing 24 of 24 books'. A table lists book records with columns for checkboxes, Cover, Title, Author, Category, ISBN, Publication, and Accession No.

☐	Cover	Title	Author	Category	ISBN	Publication	Accession No.
☐		boy	abunda	Religion	989898	lib	999-888-777
☐		sky	key	Fiction	565656	psicom	111-333-222

Figure 50. Book Page



The screenshot shows the 'Borrow/Return' page in the OPAC Admin interface. The left sidebar contains navigation links: Dashboard, Students, Books, Borrow/Return (selected), Borrowed Books, and Logout. The main content area is titled 'Borrow & Return Management' and has two tabs: 'BORROW' (selected) and 'RETURN'. Under the 'BORROW' tab, there is a 'Borrow Books' section with a 'Switch to Return Mode' button. Below this, instructions state 'Scan student ID first, then scan each book to borrow'. There are two input fields: 'Scan Student ID' with a 'Scan or type student ID' button, and 'Scan Book Barcode' with a 'Scan or type book barcode' button. Below these fields are buttons for 'Show Diagnostics', 'Test Scanner', and 'Switch to Return Mode'. At the bottom, there are sections for 'Books to Borrow' and 'Student Details'.

Figure 51. Book Borrowing Page

The screenshot shows the 'Borrow/Return' page in the OPAC Admin interface, specifically the 'RETURN' tab. The left sidebar is identical to the previous screenshot. The main content area is titled 'Borrow & Return Management' and has two tabs: 'BORROW' and 'RETURN' (selected). Under the 'RETURN' tab, there is a 'Return Books' section with a 'Switch to Borrow Mode' button. Below this, instructions state 'Scan student ID to view and return borrowed books'. There are two input fields: 'Scan Student ID (for Return)' with a 'Scan or type student ID' button, and 'Book barcode scanning disabled in return mode' with a 'Scan or type book barcode' button. Below these fields are buttons for 'Show Diagnostics', 'Test Scanner', and 'Switch to Borrow Mode'. At the bottom, there are sections for 'Currently Borrowed Books' and 'Student Details'.

Figure 52. Book Return Page



The screenshot shows the 'Borrowed Books' page in the OPAC Admin interface. The left sidebar contains the 'OPAC Admin' header with a user profile for 'admin' and a list of navigation items: Dashboard, Students, Books, Borrow/Return, and Borrowed Books (which is highlighted). A red 'Logout' button is at the bottom of the sidebar. The main content area is titled 'Borrowed Books' and includes a search bar. Below the search bar, there are two summary cards: 'Total Borrowed' with a value of 8 and 'Overdue' with a value of 7. A 'Refresh' button and a 'Calculate Fines' button are located to the right of these cards. A search bar for 'Search by student name, ID, book title, author or ISBN...' is provided. Below this is a table of currently borrowed books.

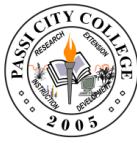
Student	Book	Borrow Date	Due Date	Status	Days Overdue	Actions
aia ID: - OQcUq_0yXUKtzhSPiz3	The Basics Of Tour Guiding Francisco M. Ramos	May 20, 2025	May 21, 2025	Overdue	3 days	View Fines Email

Figure 53. Borrowed Books Page

The screenshot shows the 'Student Borrowings' page in the OPAC Admin interface. The left sidebar is similar to the previous page but highlights 'Student Borrowings'. The main content area is titled 'Student Borrowings' and includes a search bar. Below the search bar, there is a section titled 'Students with Borrowed Books' with a 'Notify All Overdue (3)' button. A warning message states '3 students with overdue books' and 'Send notifications to remind students about their overdue books and applicable fines.' Below this is a search bar for 'Search by student name, ID, grade, section or book details...'. A list of students with overdue books is displayed, each with a profile icon, name, ID, grade, section, and number of books, along with a 'Has Overdue' status and a notification icon.

Student	Books	Status	Actions
aia ID: -OQcUq_0yXUKtzhSPiz3 • Grade: -C • Books: 1	1	Has Overdue	Notify
Aia Noel Arguelles ID: -OMjyasuC6pGjgJ7FB3 • Grade: - • Books: 3	3	Has Overdue	Notify
Aia Noel Arguelles ID: -OMk3vhXDLjvASSdFHZ_ • Grade: - • Books: 2	2	Has Overdue	Notify

Figure 54. Student Borrowings Page



**OPAC Admin**

A

admin

Admin

Borrow/Return

Borrowed Books

Student Borrowings

Announcements

Overdue Notifications

Fine Payments

Logout

Announcements

Search books, students, borrowings...

System Monitor

Settings

Send Announcements

Create and send announcements to students via email

Subject *

Message *

Attachment URL (Optional)

Link to a document or resource (optional)

Show desktop

Figure 55. Announcements Page



System Architecture

Network Model

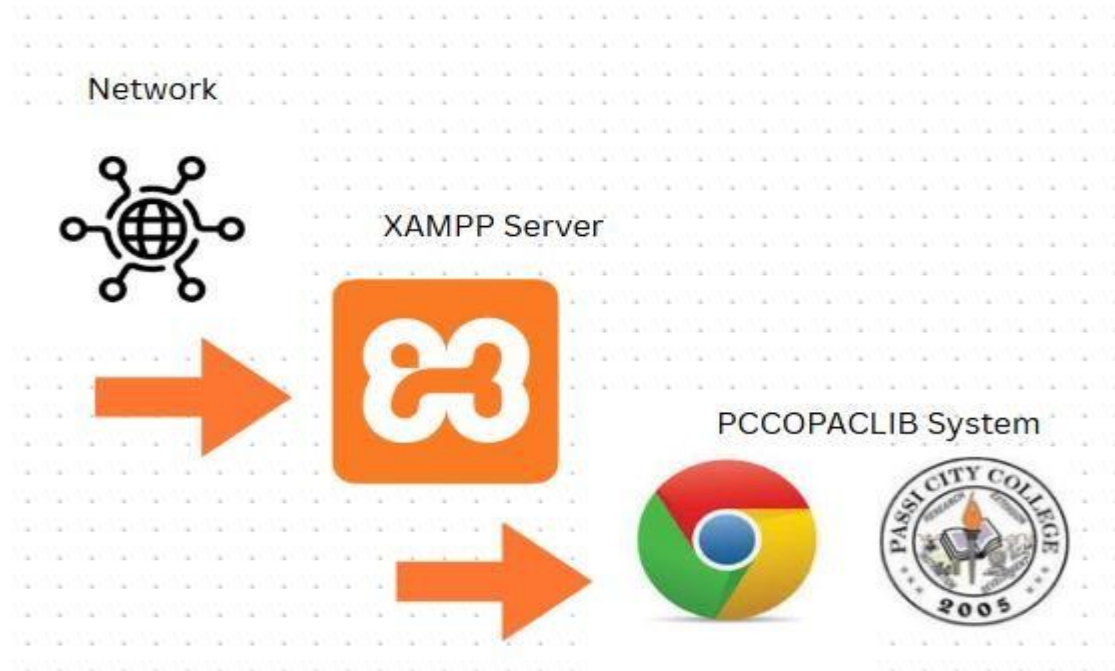


Figure 56. Network Model

Administrators and users can access the Smart Library System with Online Public Access Catalog (OPAC) through web browsers. User-submitted book and transaction data is securely stored and managed in a dedicated database server. The system utilizes the internet to enable administrators to connect to the server, manage library resources, and oversee system maintenance efficiently.



Network Topology

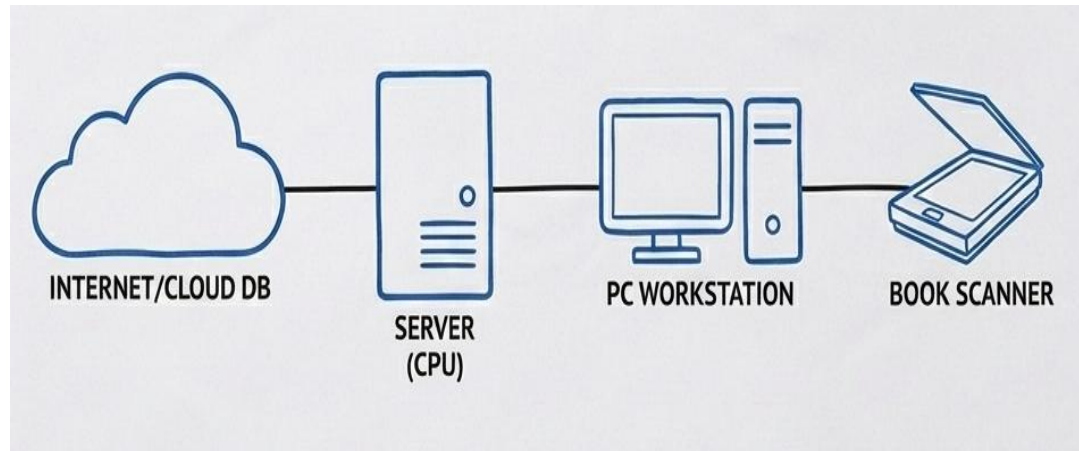
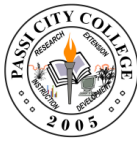


Figure 57. Network Topology

The network is designed as a linear chain, starting with an Internet connection that feeds data into a central Server acting as the system's brain. This server then processes and passes information down to a PC workstation, where the librarian or user interacts with the system. Finally, the scanner is linked to that PC, allowing physical books and student id to be digitized and saved back into the library's digital records.

Security

To uphold the confidentiality, integrity, and availability of sensitive data and the overall functionality of the Smart Library System with Online Public Access Catalog



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(OPAC) at Passi City College, the following essential security measures have been adopted:

User Authentication and Role-Based Access Control

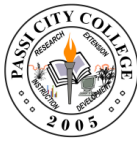
A multi-tier authentication system is in place to ensure that only authorized users can access specific system functionalities. Role-based access (RBAC) restricts actions based on user roles (e.g., administrator, librarian).

Data Encryption and Secure Transmission

Critical data, including user credentials and book records, is encrypted both in storage and during transmission using SSL/TLS protocols and AES encryption, preventing unauthorized access.

Secure Database Management

The system utilized Firebase as its Database Management System (DBMS), enforcing real-time validation, authentication-based access control, and encrypted data storage to prevent unauthorized alterations.



Regular Data Backup and Recovery Plan

Automated cloud-based backups were conducted to safeguard against accidental data loss, system crashes, or cyber threats, ensuring rapid data restoration when needed.

Network Security Protections

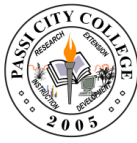
The system is secured with firewalls, Intrusion Detection Systems (IDS), and routine security updates to mitigate potential vulnerabilities and block unauthorized access.

Audit Logging and System Monitoring

A detailed logging mechanism recorded all system activities, user actions, and security events, helping to detect suspicious behavior and conduct forensic analysis in case of breaches.

Secure API Integration

If external services were connected, secure API authentication mechanisms (such as OAuth 2.0) and encrypted communication channels are employed to prevent data leaks or unauthorized access.



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Session Management and Automatic Logout

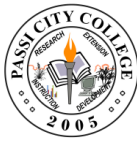
Secure session handling, including automatic session timeouts, multi-factor authentication (MFA), and token expiration policies, ensured that inactive sessions do not pose security risks.

User Awareness and Cybersecurity Training

Library staff and system users were trained on best security practices, including recognizing phishing attacks, using strong passwords, and reporting unusual activities.

Incident Response and Security Policy Compliance

A structured incident response strategy is in place to quickly contain, investigate, and resolve security breaches. Compliance with data protection regulations ensures that system security is maintained at the highest standard.



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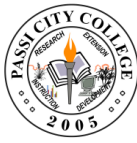
CHAPTER IV

RESULTS AND DISCUSSION

This chapter presents the results of the research study, focusing on the development, implementation, and evaluation of the Smart Library System. This highlights the effectiveness of the system in addressing the challenges associated with the efficiency and accessibility of the Passi City College Library and achieving the objectives outlined in Chapter 1. User feedback, system usability testing, and compliance with ISO/IEC 25010 standards form the basis of the findings.

The researchers successfully developed the Smart Library System using the SDLC Waterfall Methodology. The iterative design process allowed continuous feedback from stakeholders, ensuring the system met the specific requirements of Passi City College Library. The researcher deployed the system at the Passi City College Library and trained the staff members to operate and maintain it effectively.

The researchers evaluated the usability of the Smart Library System using the ISO/IEC 25010 model. The results indicate a high level of satisfaction among the one(1) office



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staff members, twenty(20) consumers, and five(5) IT Experts involved in the evaluation.

Table 7

Scales for system evaluation

Rating	Mean Range	Description
5	4.21-5.00	(E) Excellent
4	3.31-4.20	(VS) Very Satisfactory
3	2.61-3.30	(S) Satisfactory
2	1.81-2.60	(F) Fair
1	1.00-1.80	(P) Poor

Table 7 shows the scales for system evaluation to guide the respondents on how to assess the system.



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Table 8

The Functional Suitability Evaluation Result of the Respondents

Functional Suitability	Office Staff		IT Experts		Consumer		Grand Total Mean	
	Mean	Description	Mean	Description	Mean	Description	Mean	Description
Functional Completeness	4.0	VS	4.6	E	3.9	VS	4.17	VS
Functional Correctness	4.0	VS	4.8	E	3.7	VS	4.17	VS
Functional Appropriateness	4.0	VS	5.0	E	3.85	VS	4.28	E
Grand Mean	4.0	VS	4.8	E	3.82	VS	4.21	E

Table 8 summarizes the respondents' evaluation of the functional suitability of the "Smart Library System". The results indicate that office staff and consumer rated the system as "Very Satisfactory", while the IT Experts rated the system as "Excellent" across all functional suitability criteria. Specifically, the system achieved high marks for Functional Completeness (Mean: 3.9-4.6), demonstrating that it effectively covers all specified tasks and objectives. It also excels in Functional Correctness (Mean: 3.7-4.8), consistently delivering accurate results with the required precision. Moreover, the system demonstrates strong



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Functional Appropriateness (Mean: 3.8-5), effectively supporting the accomplishment of tasks and objectives.

The overall Grand Mean of 4.21 confirms that the system is highly effective and reliable across all user categories—Office Staff, IT Experts, and Consumers. These results suggest that the system is well-designed, exceeds user expectations, and delivers outstanding functionality.

Table 9

The Performance Efficiency Evaluation Result of the Respondents

Performance Efficiency	Office Staff		IT Experts		Consumer		Grand Total Mean	
	Mean	Description	Mean	Description	Mean	Description	Mean	Description
Time Behavior	4.0	VS	4.8	E	3.55	VS	4.12	VS
Resource Utilization	4.0	VS	4.6	E	3.45	VS	4.02	VS
Capacity	4.0	VS	5.0	E	3.55	VS	4.18	VS
Grand Mean	4.0	VS	4.8	E	3.52	VS	4.11	VS

Based on the evaluation presented in Table 9, the office staff, IT experts, and consumers assessed the Smart Library System. The system's Time Behavior, which measures response



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time, processing speed, and throughput, received an "Excellent" rating from IT experts (4.8), while the office staff (4) and consumers (3.55) rated it "Very Satisfactory". With an overall mean of 4.12, this criterion was categorized as "Very Satisfactory".

For Resource Utilization, which assesses the efficiency of the system's resources, It Experts gave the highest rating of 4.6, followed by Office Staff with 4 and IT Experts with 3.45. The overall mean for this criterion was 4.02, rated as "Very Satisfactory".

The system's capacity, which evaluates its ability to meet maximum operational limits, was rated "Excellent" by the IT Experts with the score of 5, while Office Staff and Consumers rated it as "Very Satisfactory" with the score of 4 and 3.55 respectively. This criterion received the highest overall mean rating of 4.18, categorized as "Very Satisfactory."

The Grand Mean across all criteria and groups was 4.11, which is classified as "Very Satisfactory."

The results indicate that the system performs well in all performance efficiency criteria. Users across all groups



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expressed satisfaction with its Time Behavior, Resource Utilization, and Capacity. Despite slight score variations, the system consistently meets or exceeds performance requirements, demonstrating reliability and effectiveness.

Table 10

The Compatibility Evaluation Result of the Respondents

Compatibility	Office Staff		IT Experts		Consumer		Grand Total Mean	
	Mean	Description	Mean	Description	Mean	Description	Mean	Description
Co-existence	3.0	S	4.2	VS	3.45	VS	3.55	VS
Interoperability	3.0	S	5.0	E	3.75	VS	3.92	VS
Grand Mean	3.0	S	4.6	E	3.6	VS	3.73	VS

The evaluation of the compatibility criteria reveals that the system's co-existence functionality achieved an overall "Very Satisfactory" rating. IT Experts assigned the highest mean score of 4.2, followed by Consumers with 3.45, and office staff with 3, which falls under the "Satisfactory" category. Likewise, the system's interoperability received an overall "Very Satisfactory" rating, with a grand mean of 3.92.



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IT experts scores 5.0, while consumers scores 3.75 and office staff rated it 3 ("Satisfactory").

The system demonstrates strong compatibility, with excellent performance in coexisting with other products and very satisfactory interoperability. While consumer feedback shows the highest satisfaction, there is room for a slight improvement in meeting office staff expectations. Overall, the system meets its compatibility criteria effectively, with a grand mean of 3.73 ("Very Satisfactory").

Table 11

The Usability Evaluation Result of the Respondents

Usability	Office Staff		IT Experts		Consumer		Grand Total Mean	
	Mean	Description	Mean	Description	Mean	Description	Mean	Description
Appropriateness Recognizability	4.0	VS	4.8	E	4.1	VS	4.3	E
Learnability	4.0	VS	4.8	E	3.95	VS	4.25	E
Operability	4.0	VS	5.0	E	4.0	VS	4.33	E
User Error Protection	4.0	VS	4.6	E	3.55	VS	4.05	VS
User Interaction Aesthetics	3.0	S	4.8	E	3.8	VS	3.87	VS
Accessibility	4.0	VS	5.0	E	4.1	VS	4.37	E



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Grand Mean	3.83	VS	4.83	E	3.92	VS	4.19	VS
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The usability evaluation of the Smart Library System revealed an overall "Very Satisfactory" rating, with a grand mean of 4.19 across various user groups, including Office Staff, IT Experts, and Consumers. The system received an "Excellent" rating in several key areas. Appropriateness Recognizability (4.3) ensures users can easily determine its relevance to their needs. Learnability (4.25) indicates that the system is easy to learn and use efficiently. Operability (4.33) highlights its ease of operation and control. Lastly, Accessibility (4.37) demonstrates strong usability for a wide range of users. However, User Error Protection received a slightly lower rating of 4.05 ("Very Satisfactory"), and User Interaction Aesthetics (3.87). Indicating that while adequate error prevention measures are in place and reflects a visually pleasing interface, there is room for improvement.

The system has been evaluated as highly usable across all groups, with most criteria rated "Excellent." The highest-rated aspect is Accessibility, indicating that the system is particularly easy to access to use it efficiently. However, User Error Protection and User Interaction



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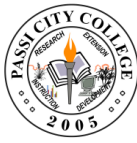
Aesthetics received a slightly lower "Very Satisfactory" score, suggesting an area for potential improvement. Overall, the system meets and exceeds expectations for usability, making it suitable for a diverse range of users.

Table 12

The Reliability Evaluation Result of the Respondents

Reliability	Office Staff		IT Experts		Consumer		Grand Total Mean	
	Mean	Description	Mean	Description	Mean	Description	Mean	Description
Maturity	4.0	VS	4.6	E	3.65	VS	4.08	VS
Availability	4.0	VS	4.4	E	3.9	VS	4.1	VS
Fault Tolerance	4.0	VS	5.0	E	3.65	VS	4.2	VS
Grand Mean	4.0	VS	4.67	E	3.73	VS	4.13	VS

Based on the reliability criteria, the system demonstrated strong performance across all groups surveyed. For Maturity, the system received "Excellent" ratings from IT Experts (4.6). However, the system received "Very Satisfactory" from Office Staff (4), and Consumers (3.65), with a grand mean of 4.08 ("Very Satisfactory"). In terms of



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Availability, IT Experts rated it as "Excellent" (4.4), while Office Staff and Consumers rated it "Very Satisfactory" (4.00 and 3.9, respectively), leading to an overall grand mean of 4.1 ("Very Satisfactory"). For Fault Tolerance, IT Experts rated the system as "Excellent," with the score of 5, while Office Staff scores 4, and Consumers 3.65 rated as "Very Satisfactory", resulting in a grand mean of 4.2 ("Very Satisfactory").

The system consistently meets the reliability criteria, receiving predominantly "Very Satisfactory" ratings across Maturity, Availability, and Fault Tolerance. This indicates that it performs reliably under normal operations, remains accessible when required, and functions as intended, even in the presence of hardware or software faults. Overall, the system is reliable, with a grand mean of 4.13, reflecting an "Very Satisfactory" level of performance.



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Table 13

The Security Evaluation Result of the Respondents

Security	Office Staff		IT Experts		Consumer		Grand Total Mean	
	Mean	Description	Mean	Description	Mean	Description	Mean	Description
Confidentiality	4.0	VS	4.6	E	3.6	VS	4.07	VS
Integrity	4.0	VS	5.0	E	3.6	VS	4.2	VS
Non-repudiation	4.0	VS	5.0	E	4.85	E	4.67	E
Grand Mean	4.0	VS	4.87	E	4.02	VS	4.30	E

The evaluation of the system's security criteria, based on feedback from various user groups, resulted in an overall "Excellent" rating with a grand mean of 4.30. The system excelled in Non-repudiation (4.67), ensuring data protection against unauthorized access and providing verifiable proof of actions taken. Confidentiality and Integrity received a slightly lower but still "Very Satisfactory" rating of 4.07 and 4.2 respectively, indicating that while data access is restricted to authorized users and adherence to ethical principle, there is room for enhancement. Overall, the system demonstrates robust security features, with consistently high



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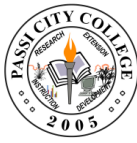
ratings affirming its reliability and effectiveness in safeguarding data.

Table 14

The Maintainability Evaluation Result of the Respondents

Maintainability	Office Staff		IT Experts		Consumer		Grand Total Mean	
	Mean	Description	Mean	Description	Mean	Description	Mean	Description
Analyzability	4.0	VS	5.0	E	3.95	VS	4.32	E
Changeability	4.0	VS	4.6	E	3.45	VS	4.02	VS
Stability	3.0	S	4.2	VS	3.75	VS	3.65	VS
Testability	3.0	S	5.0	E	4.2	VS	4.07	VS
Grand Mean	3.5	VS	4.7	E	3.84	VS	4.01	VS

The assessment of the system's maintainability considered four criteria: analyzability, changeability, stability, and testability. The results ranged from "Very Satisfactory" to "Excellent." Analyzability received a grand mean of 4.32 ("Excellent"), with IT experts rating it "Excellent" (5), while office staff (4) and consumers (3.95) rated it "Very Satisfactory". IT Experts rated Changeability



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as "Excellent" with a score of 4.6, while Office staff and Consumers gave it a 4 and 3.45. These ratings resulted in a grand mean of 4.02. Stability achieved an "Very Satisfactory" rating across IT Experts and Consumers with a score of 4.2 and 3.75 respectively, with office staff (3) rated as "Satisfactory", leading to a grand mean of 3.65 rated as "Very Satisfactory". Testability also received high ratings, with IT Experts marking it "Excellent" (5), while Consumers (4.2) rated it "Very Satisfactory" and Office Staff (3) rated it "Satisfactory", resulting in an overall mean of 4.07. These findings indicate strong system maintainability, particularly in stability and testability.

The system maintains a high standard across all criteria, as reflected in the overall grand mean of 4.01, categorized as "Very Satisfactory." The Analyzability aspects received exceptionally high ratings, highlighting the system's ease of understanding its inner workings. Although Changeability, Stability and Testability received slightly lower ratings, they still achieved a "Very Satisfactory" score, demonstrating the system's strong overall performance in maintainability.



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Table 15

The Portability Evaluation Result of the Respondents

Portability	Office Staff		IT Experts		Consumer		Grand Total Mean	
	Mean	Description	Mean	Description	Mean	Description	Mean	Description
Adaptability	3.0	S	5.0	E	4.2	VS	4.07	VS
Installability	3.0	S	5.0	E	3.95	VS	3.98	VS
Grand Mean	3.0	S	5.0	E	4.08	VS	4.03	VS

The results for the portability criteria indicate that the system demonstrates strong adaptability and installability across all user groups. Specifically, adaptability received a mean score of 5 (Excellent) from IT Experts, 4.2 (Very Satisfactory) from Consumers, and 3 (Satisfactory) from Office Staff, with a grand mean of 4.07 (Very Satisfactory). Similarly, installability was rated as 5 (Excellent) by IT Experts, 3.95 (Very Satisfactory) by Consumers, and 3 (Satisfactory) by Office Staff, resulting in a grand mean of 3.98 (Very Satisfactory).

The highly portable system is adaptable and installable across varying user environments. This highlights its



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effectiveness and efficiency in meeting diverse operational and usage requirements.

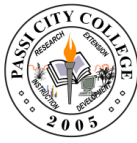
Table 16

Summary Evaluation Results Evaluated by the IT Experts

ISO 25010 Criteria	Mean	Interpretation
Functional Suitability	4.8	E
Performance Efficiency	4.8	E
Compatibility	4.6	E
Usability	4.83	E
Reliability	4.67	E
Security	4.87	E
Maintainability	4.7	E
Portability	5.0	E
Over-all Mean	4.78	E

The evaluation of the IT system using the ISO 25010 criteria shows outstanding performance across all aspects. The criteria, all received "Excellent" ratings ranging from 4.6 to 5.

The IT system demonstrates exceptional quality in key areas such as functionality, performance, reliability, and



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security. The overall mean score of 4.78 confirms the system's excellence.

Table 17

Summary Evaluation Results Evaluated by the Consumer

ISO 25010 Criteria	Mean	Interpretation
Functional Suitability	3.82	VS
Performance Efficiency	3.52	VS
Compatibility	3.6	VS
Usability	3.92	VS
Reliability	3.73	VS
Security	4.02	VS
Maintainability	3.84	VS
Portability	4.08	VS
Over-all Mean	3.82	VS

The evaluation results based on the ISO 25010 criteria indicate a strong overall performance. The system received Very Satisfactory ratings across all criteria. The overall mean score is 3.82, which reflects a generally high level of satisfaction among consumers.



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The system performs excellently in almost all key areas. This suggests that it is a robust and reliable system that meets the needs and expectations of its users.

Table 18

Summary of Over-all Evaluation Results

ISO 25010 Criteria	Mean	Interpretation
Functional Suitability	4.21	E
Performance Efficiency	4.11	VS
Compatibility	3.73	VS
Usability	4.19	VS
Reliability	4.13	VS
Security	4.30	E
Maintainability	4.01	VS
Portability	4.03	VS
Over-all Mean	4.09	VS

The evaluation results based on the ISO 25010 criteria show that the system performed exceptionally well across most categories. The system achieved an excellent rating in Functional Suitability (4.21), and Security (4.30) while Performance Efficiency (4.11), Compatibility (3.73) Usability (4.19), Reliability (4.13), Maintainability (4.01), and



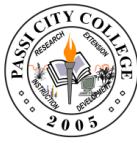
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Portability (4.03) received a "Very Satisfactory" rating. The Overall Mean score of 4.09 signifies a very satisfactory performance overall.

The system demonstrates high quality and reliability, with excellent performance across most criteria. Yet minor improvements could increase satisfaction even further.

The "Smart Library System" provides positive results in its performance in various quality and system characteristics, including functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. It addressed the inefficiencies of manual process of borrowing and return in the PCC Library. By integrating modern technology, the system streamlined operations, reduced errors, and enhanced user communication. Moreover, its compliance with ISO standards ensures its robustness and reliability, making it a sustainable solution for water resource management.



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CHAPTER V

SUMMARY, FINDING, CONCLUSIONS, AND RECOMMENDATIONS

Summary

The main objective of the study was to determine the acceptability of the system in terms of its usability. The Smart Library System was a program developed using HTML and Python for its front-end and back-end, with Firebase serving as its database. It was designed for Passi City College Library to facilitate easier and faster transactions, such as searching the library catalog, managing book borrowing and returns, and generating student library IDs. The user-friendly environment enabled users to perform tasks and commands through mouse click events. The Smart Library System was implemented using a client-server connection.

After development, the system was tested and evaluated for its usability. This evaluation was conducted by distributing the ISO/IEC 25010 model to the respondents. The collected data were analyzed using standard scales for system evaluation. The study was conducted at Passi City College, City of Passi, Iloilo.



Findings

The findings of the study revealed that:

1. The Smart Library System was developed following the SDLC Waterfall Methodology Model.
2. The system obtained an overall mean rating of 4.09 based on the ISO/IEC 25010 evaluation, which was described as "Very Satisfactory." The system scored exceptionally high in Functional Suitability (4.21) and Security (4.30), both rated as "Excellent," and was considered highly acceptable by the end users.

Conclusions

Based on the results and findings, the researchers concluded that the Smart Library System was efficient and effective for use, as indicated by the "Very Satisfactory" overall rating from the evaluation. The study demonstrated that the system could address the challenges of manual library processes by providing a comprehensive and secure solution capable of handling large volumes of data while ensuring fast and efficient access to records.

The findings highlighted the importance of the Smart Library System for Passi City College in enabling the institution to effectively manage library data, reduce manual errors, and enhance user communication, thereby providing a



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complete solution for the administration of library services. Overall, the proposed system demonstrated significant advantages, and the shift from manual to automated borrowing and returning processes proved beneficial to PCC personnel, staff, and users.

Recommendations

The following recommendations were formulated based on the study's findings:

1. The researchers recommended that PCC personnel, particularly the library staff at Passi City College, consider implementing the proposed Smart Library System. Its adoption would facilitate faster, more efficient, and secure access to library records and operations, leveraging the system's high ratings in Functional Suitability and Security, as well as its "Very Satisfactory" rating in Performance Efficiency, to enhance overall productivity and data management capabilities.
2. Future researchers conducting similar studies may consider improving, adding more features to, and updating the proposed system, particularly in the areas of Compatibility (3.73) and Usability (4.19), to further enhance the user experience.



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APPENDICES

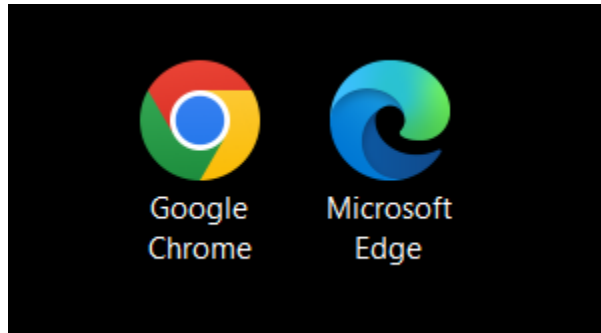
Appendix A

User's Manual

The "Smart Library System" application was developed to replace the manual process of managing library operations at Passi City College. Its primary purpose was to provide a user-friendly, web-based platform designed for desktop computers to streamline library services.

The system aimed to improve the cataloging, borrowing, and returning of books at the Passi City College Library. Its goal was to make library transactions faster, more secure, and more efficient by incorporating advanced web features into the desktop platform, ensuring the preservation and easy accessibility of library information. To access the local host, users were instructed to follow the provided steps to launch the application.

1. Open the Web Browser.



2. Types the IP address of the main server and name of the application folder in the address bar.
(<http://library.passicitycollege-pcc.com>)

3. After that just click enter to proceed in the login form then type the username and password for user/admin.

4. The admin dashboard appears showing the total number of Books, Students, Books Borrowed, and Overdue Books.

5. If you scroll down you can see the Library Usage. Also the Quick Actions for Borrowing/Return, Add Books, Manage Students, and Send Announcement.

6. When you click the "Student" in the navigation bar it will show another page for the students' management. It show the Generated Student IDs, Print Selected Barcodes, Add New Student Button and list of added Student with their details where you can view, edit, and delete them.

6.1. Generated Student IDs - it will show a dialog box where you can see the student details with actions of



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view and print or you can select all or selected students to print their IDs.

6.2. Print Selected Barcode- If students' only need their barcode, just select a student from the inventory and click to select and to print the barcode.

6.3. Adding New Student - Click Add Student and it will show a dialog box where you can see details needed to add a student. After filling up, just click Add student button.

6.4. Updating student information. Select a student from inventory. Click Edit. Update the necessary fields. Update.

7. When you click the "Books" in the navigation bar it will show another page for the book management. It show the Print Barcode and Add New Book and below are the list of added books with its details where you can also view, edit or delete them.

7.1. Print Barcode- upon clicking a dialog box will appear. It shows the book details and you can select all or selected books to print their barcode.

7.2. Add New Books- if click it shows the details need to be filled up. Enter details and Add.



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7.3. Updating book information. Select a book from inventory. Click Edit. Update the necessary fields. Update.

8. When you click on "Borrow/Return" in the navigation bar of the dashboard, it displays a new page dedicated to Borrow/Return management. On this page, you can borrow/return a book by scanning only the barcode from their generated IDs and the book they want to borrow. Check details and confirm request by clicking borrow book/s.

8.1. When returning, scan student barcode and it will show books borrowed by students. Confirm details and return,

9. When you click on "Student Borrowings" in the navigation bar of the dashboard, it displays a new page dedicated to Student with Borrowed Books. On this page, you can view the student with an active borrow.

10. When you click on "Reports" in the navigation bar of the dashboard, it displays a new page dedicated to reports management. On this page, you see the reports the system can generate.

11. When you click on "Announcements" in the navigation bar of the dashboard, it displays a new page dedicated to Announcements or important reminders from the admin and send to selected or all of the students registered.



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12. When you click on "Overdue Notifications" in the navigation bar of the dashboard, it displays a new page dedicated to overdue students. On this page, you can view the details of the student, number of days overdue and its fines. And it will also notify students who have an overdue book.

13. When you click on "Fine Payments" in the navigation bar of the dashboard, it displays a new page dedicated to Fine Payment Inventory. On this page, you can view the complete details of each student who paid their fines and its status. You can also print it.

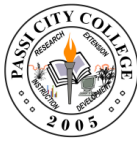
14. When you click on "Attendance" in the navigation bar of the dashboard, it displays a new page dedicated to Record Attendance and List&Print attendance records.

14.1. Record Attendance- scan student id to log in and log out.

14.2. List & Print- view and print attendance records.

15. When you click on "User Management" in the navigation bar of the dashboard, it displays a new page dedicated to User management. On this page, you can add or edit users and can manage permissions.

16.If you scroll in the navigation bar, you can see the system monitor to monitor if the system is running well.



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17. the view generated id if click shows student details and actions view & print student IDs.

18. print book barcode upon clicking show the book details and actions to print it.

19. Library Settings if click, appears a dialog box and shows book borrowing rules. Can be updated and if done just save changes.

20. This is the user account feature where you can view the total books of the library. You can search books and see its availability.



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City of Passi, Iloilo



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Appendix B

Approved Letter to Conduct the Study

February 17, 2025

EDMAR T. LABRADOR, R.L. MLIS
College Librarian
Passi City College
City of Passi, Iloilo

Dear Mr. Labrador,

Greetings of peace!

Capstone is an integral part of academic institutions, and it plays a vital role in advancing knowledge and driving innovation.

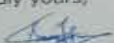
We, the Bachelor of Science in Information Technology (BSIT) student wish to conduct our Capstone Project entitled "**Smart Library System**". This study aims to develop a web-based system integrated with an Online Public Access Catalog (OPAC) for Passi City College. The main goal of this system is to streamline the borrowing and returning of books without so much hassle.

In this connection, we would like to ask permission from your good office to allow us to conduct the study this second semester of academic year 2024-2025.

Furthermore, your approval is sought to make this plan a reality. More than your authorization is a manifestation of your strong support in the college.

Thank you for your unwavering support to continuously empower us, the students.

Truly yours,


KRISHA LYNE P. LAGO
Leader, Capstone Project

Noted:


RODELIN A. DE JUAN
Adviser, Capstone Project

Recommending Approval:


ENGR. MYLEN B. PADERES, MEng'g
Dean, SOICT

Approved:


EDMAR T. LABRADOR, R.L. MLIS
College Librarian



Appendix C

Approved Letter for Thesis Adviser

February 17, 2025

MR. RODELIN A. DE JUAN
SOICT Instructor
Passi City College
City of Passi, Iloilo

Sir:

Warm Greetings!

We humbly request your permission to serve as capstone project adviser. Given your experience and dedication to Information Technology, we are confident that your guidance and mentorship will be invaluable to the successful completion of our capstone project.

The proposed capstone project is entitled "**Smart Library System**". With your guidance, we are confident that we can develop a comprehensive and meaningful capstone project that contributes to the existing body of knowledge in this field. We fully understand the significant time and effort required for this role and genuinely appreciate your consideration in taking on this responsibility.

We are enthusiastic about the opportunity to work under your mentorship and sincerely hope for positive response and support in our academic endeavor.

Respectfully yours,


KRISHA LYNE P. LAGO
Leader, Capstone Project

Noted:


JERAHMEEL P. CANTOS
Capstone, Instructor

Recommending Approval:


ENGR. MYLEN B. PADERES, MEng'g
Dean, SOICT

Approved:


RODELIN A. DE JUAN
SOICT Instructor



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City of Passi, Iloilo



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Appendix D

Letter For Deployment

Republic of the Philippines
PASSI CITY COLLEGE
City of Passi, Iloilo
passicitycollege@gmail.com
Passi City College-PCCL
033-311-4931

SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

November 7, 2025

EDMAR T. LABRADOR, R.L. MLIS
College Librarian
Passi City College
City of Passi, Iloilo

Dear Mr. Labrador:

Greetings of peace and excellence!

We, the Fourth-Year Bachelor of Science in Information Technology students of Passi City College, respectfully submit this letter to request approval for the deployment of our Capstone Project, entitled **"Smart Library System"**.

This project is a requirement in partial fulfillment of our degree program, aiming to apply the knowledge and skills we have gained throughout our academic journey. The deployment stage will allow us to evaluate the system's functionality, effectiveness, and reliability in an actual or simulated environment.

We assure you that the deployment process will be carried out responsibly and ethically, following the school's guidelines and standards. Furthermore, all data gathered during the process will be used strictly for academic and research purposes.

We hope for your kind approval and support in allowing us to proceed with the deployment of our project. Should you require further information, we are willing to provide additional documentation or a presentation as needed.

Thank you very much for your continued guidance and support.

Respectfully yours,

KRISHA EYNE P. LAGO
Capstone Project Leader

JHARAYA CAMILLE L. LABIAGA
Capstone Project Team

RUSSEL H. LASAP, JR.
Capstone Project Team

WILSON L. JAGOLINGAY, JR.
Capstone Project Team

CHESTER D. PORRAS
Capstone Project Team

Noted by:

RODELINA A. DE JUAN
Capstone Project Advisor

Endorsed by:

ENGR. MYLEN B. PADARES, MEng'g
Dean, SOICT

Approved:

EDMAR T. LABRADOR, R.L. MLIS
College Librarian
Passi City College

MISSION
Passi City College is committed to provide quality education and training with the help of modern technology and human resources to develop the students' skills and knowledge for their future.

VISION
To become the best in the field of education and training, providing quality education and training to the students of the future.



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City of Passi, Iloilo



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Appendix E

Approved Letter for English Critique

Republic of the Philippines
PASSI CITY COLLEGE
City of Passi, Iloilo
passicitycollege@gmail.com
Passi City College-PCO
033-311-6087

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December 3, 2025

RHODORA P. PANIZAL, Ed.D.
Faculty, SOTE
Passi City College
City of Passi, Iloilo

Dear Dr. Panizal:

Good day!

We respectfully write to request your assistance in serving as the English Critic for our Capstone Project manuscript titled **"Smart Library System"**, which is one of the requirements for our degree, Bachelor of Science in Information Technology.

As we work toward completing our documentation, we aim to ensure that our paper meets the standards of clear, coherent, and grammatically accurate academic writing. With your expertise in English and technical writing, we believe that your guidance will greatly help us refine our manuscript and improve its overall quality.

We truly appreciate your time and consideration. Your support would be a valuable contribution to the successful completion of our capstone project.

Thank you very much, and we look forward to your positive response.

Respectfully yours,

KRISHA LYNE LAGO
Capstone Project Leader

JHARAIA CAMILLE LABIAGA
Capstone Project Team

WILSON JAGOLINGAY, JR.
Capstone Project Team

RUSSEL LASAP, JR.
Capstone Project Team

CHESTER PORRAS
Capstone Project Team

Noted:

RODELINA A. DE JUAN
Capstone Project Adviser

Endorsed by:

ENGR. MYLEN B. PADERES, MEng'g
Dean, SOICT

Approved:

RHODORA P. PANIZAL, Ed.D.
Faculty, SOTE

12/11/25

VISION
Passi City College: A Center of Academic Excellence for Sustainable Development and Societal Transformation.

MISSION
Passi City College is committed to produce a globally competent graduates who are well-equipped with relevant academic and technological skills and values which enable them to become productive citizens and collaborators for socio change.

COLLEGE GOALS & OBJECTIVES
To Accomplish this Mission, Passi City College Commits itself to:
• Equip students with adequate and relevant knowledge and skills through technology-oriented education.
• Provide instruction, research, and community extension.
• Respect for human dignity, preserve local and national identity.
• Promote peace and service to the community and contribute to the acceleration of the country's economic growth and development and
• Bring them close to God and mold them to lead quality life in a democratic, just, peaceful, enrichment-friendly and God-loving society.



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City of Passi, Iloilo

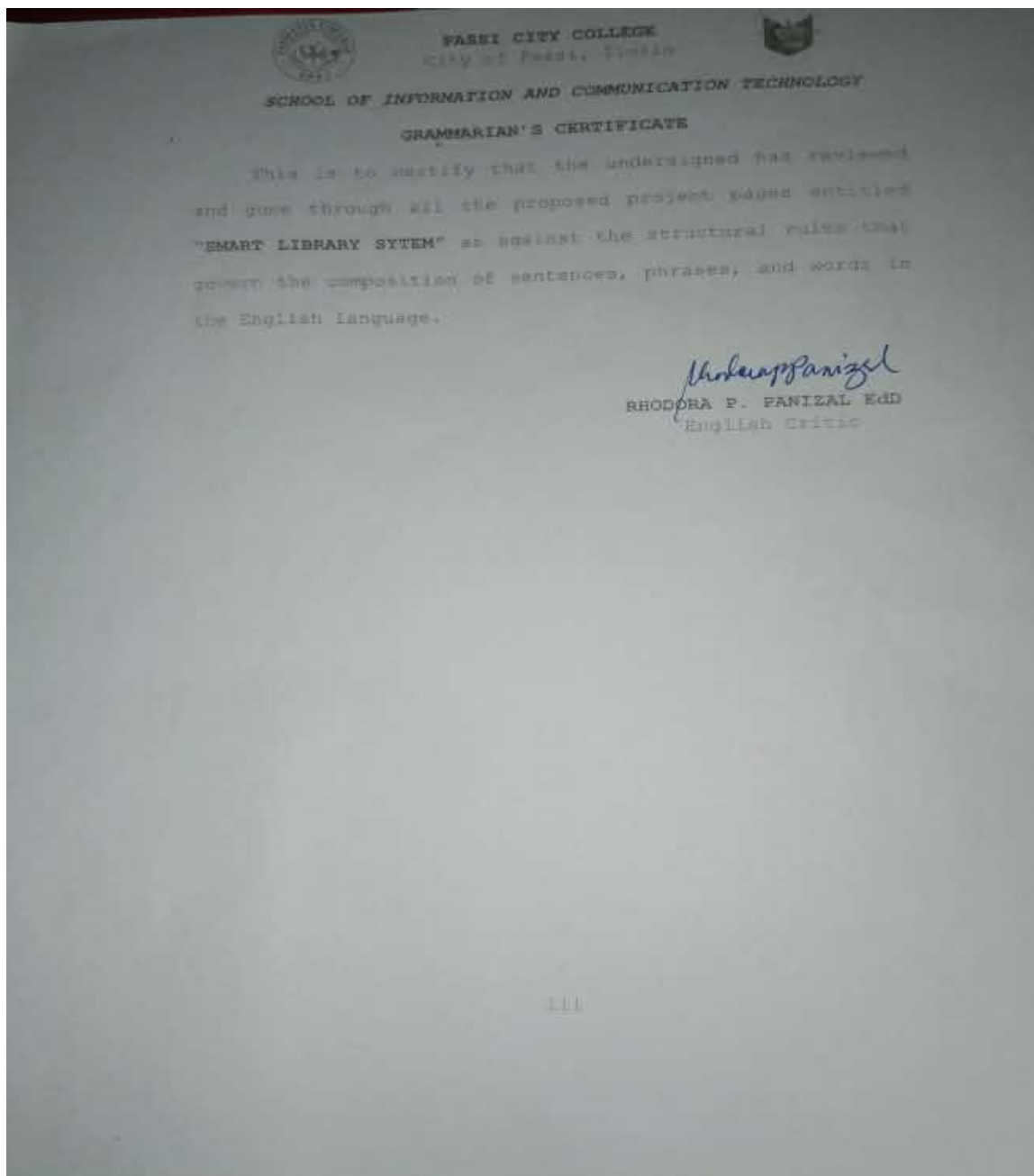


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Appendix F

Grammarian Certification





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Appendix G

Research Instrument

Evaluation Sheet

SMART LIBRARY SYSTEM

This Survey form is being conducted by Krisha Lyne Lago, Jharaia Camille Labiaga, Wilson Jagolingay, Jr., Russel Lasap, and Chester Porras as part of the requirements for a degree Bachelor of Science in Information Technology at Passi City College. The purpose of this questionnaire is to evaluate and validate the system in terms of its product and quality in use as ISO/IEC 25010 standard for the evaluation of system and software quality.

Instruction:

Kindly fill up this form appropriately. Please don't leave unanswered question. Your answer will remain strictly confidential, neither your name nor your individual responses will be given to any individual or organization either inside nor outside Passi City College.

Your cooperation and honesty are greatly appreciated.

Part I. Respondent's Profile

Name of Evaluator(optional): _____

- ☐ Office Staff
☐ IT Expert
☐ Consumer

Part II. Assessment of the system based on ISO/IEC 25010 quality standards.

Direction: Please evaluate and validate in terms of its Over-all Performance by marking / checking the corresponding box which best describe the statement.

Legend:

	Numerical Value	Description
1	Poor (P)	
2	Fair (F)	
3	Satisfactory (S)	
4	Very Satisfactory (VS)	
5	Excellent (E)	



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Criteria	1	2	3	4	5
Functionality Suitability					
Functional Completeness The system covers all the specified tasks and user objectives.					
Functional Correctness The system provides the correct results with the needed degree of precision.					
Functional Appropriateness It facilitates the accomplishment of specified tasks and objectives.					
Performance Efficiency	1	2	3	4	5
Time Behavior The system response and processing times and throughput rates when performing its functions, meet requirements.					
Resource Utilization The system amounts and types of resources used when performing its functions, meet requirements.					
Capacity The system maximum limits of parameter meet requirements.					
Compatibility	1	2	3	4	5
Co-existence The system can perform its required functions efficiently while sharing a common environment and resources with other products, without detrimental impact on any other product.					



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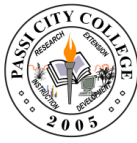
Interoperability • The system can exchange information and use the information that has been exchanged.					
Usability	1	2	3	4	5
Appropriateness Recognizability • The system allows users to recognize if it is appropriate for their needs.					
Learnability • The system can be used by specified users to achieve specified goals of learning to use the application with effectiveness, efficiency, freedom from risk and satisfaction in a specified context of use.					
Operability • The system has attributes that make it easy to operate and control.					
User Error Protection • The system protects user against making errors.					
User Interaction Aesthetics • The user interface enables pleasing and satisfying interaction for the user.					
Accessibility • It can be used by people with the widest range of characteristics and capabilities to achieve a specified goal in a specified context of use					
Reliability	1	2	3	4	5
Maturity • The system meets the needs for reliability under normal operation.					
Availability • It is operational and accessible when required for use.					
Fault Tolerance					



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• It operates as intended despite the presence of hardware or software faults.					
Security	1	2	3	4	5
Confidentiality • The system ensures that data are accessible only to those authorized to have access.					
Integrity • The system prevents unauthorized access to, or modification of, computer programs or data.					
Authenticity • It can be proven to which the identity of a subject or resource can be proved to be the one claimed.					
Maintainability	1	2	3	4	5
Analyzability Faults be easily diagnosed.					
Changeability • The system can be easily modified.					
Stability • It can continue functioning if changes are made.					
Testability • The system can be easily tested.					
Portability	1	2	3	4	5
Adaptability • The system can effectively and efficiently be adapted for different or evolving hardware, software or other operational or usage environment					



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Installability

- The effectiveness and efficiency of a system that can be successfully installed and/or uninstalled in a specified environment.

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Recommendations / Suggestions:

Signature



Appendix H

Weekly Meeting



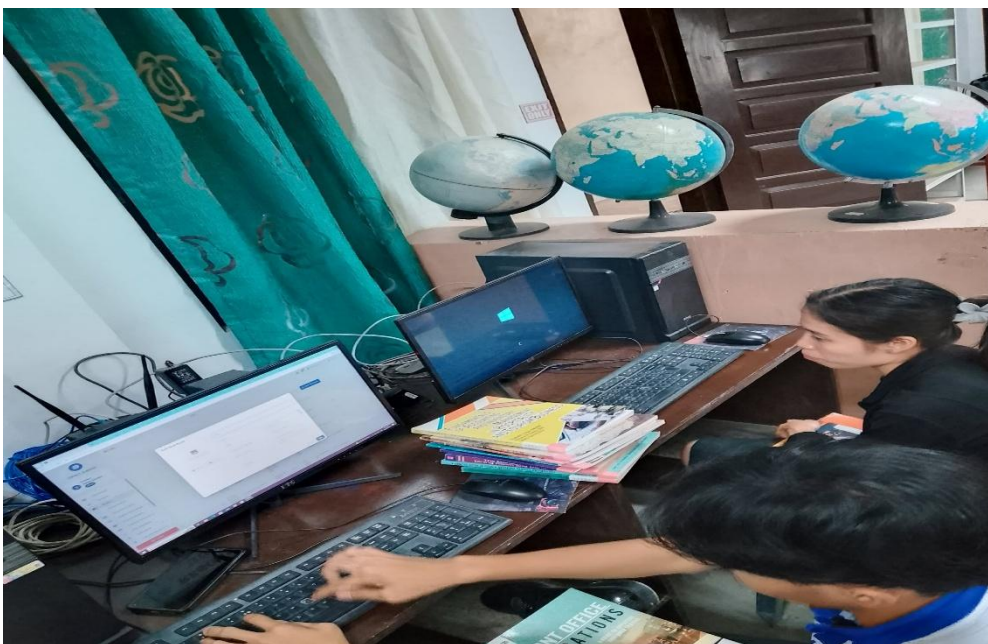


The Researchers conducted a weekly meeting.



Appendix I

Testing of the Developed System





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City of Passi, Iloilo



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Appendix J

Evaluation of the Developed System



The Researchers conducted a System Evaluation with
Passi City College SOICT Instructors and Students.



Appendix K

Personal Data

Name : Krisha Lyne P. Lago
Home Address : Bugtongan Dueñas, Iloilo
Birthdate : July 02, 2003
Gender : Female
Civil Status : Single
Citizenship : Filipino
Fathers Name : Jose Eran T. Lago
Mothers Name : Elsie P. Lago



Educational Background

Elementary : Bugtongan Elementary School
Bugtongan Dueñas, Iloilo
2010-2016
Secondary : Sisters of Mary School-Girlstown, Inc.
Poblacion City of Talisay, Cebu
2016-2021
Tertiary : Passi City College
City of Passi, Iloilo
Degree : Bachelor of Science in Information
Technology



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Personal Data

Name : Jharaia Camille Labiaga
Home Address : Punong, Passi City
Birthdate : June 24, 2004
Gender : Female
Civil Status : Single
Citizenship : Filipino
Fathers Name : Jose Reno Labiaga
Mothers Name : Elsie Labiaga



Educational Background

Elementary : Dueñas Central Elementary School
Dueñas, Iloilo
2010-2016
Secondary : Dueñas General Comprehensive High
School
Dueñas, Iloilo
2016-2021
Tertiary : Passi City College
City of Passi, Iloilo
Degree : Bachelor of Science in Information
Technology



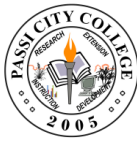
Personal Data

Name : Wilson L. Jagolingay, Jr.
Home Address : Abangay Pototan, Iloilo
Birthdate : January 09, 2003
Gender : Male
Civil Status : Single
Citizenship : Filipino
Fathers Name : Wilson N. Jagolingay, Sr.
Mothers Name : Marina L. Jagolingay



Educational Background

Elementary : Batuan Elementary School
Batuan Pototan, Iloilo
2010-2016
Secondary : Pototan National Highschool
Cau-yan Pototan, Iloilo
2016-2022
Tertiary : Passi City College
City of Passi, Iloilo
Degree Earned : Bachelor of Science in Information
Technology



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Personal Data

Name : Russel H. Lasap, Jr.
Home Address : Gibato Dumarao, Capiz
Birthdate : June 20, 2004
Gender : Male
Civil Status : Single
Citizenship : Filipino
Fathers Name : Russel V. Lasap, Sr.
Mothers Name : Daisy H. Lasap



Educational Background

Elementary : Gibato Elementary School
Gibato Dumarao, Capiz
2010-2016
Secondary : Ramon A. Benjamin Sr. National
High School
Dacuton Dumarao Capiz
2016-2022
Tertiary : Passi City College
City of Passi, Iloilo
Degree Earned : Bachelor of Science in Information
Technology



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Personal Data

Name : Chester J. Porras
Home Address : Aglanot, Dumarao, Capiz
Birthdate : February 19, 2002
Gender : Male
Civil Status : Single
Citizenship : Filipino
Fathers Name : Graciano D. Porras
Mothers Name : Neddy J. Porras



Educational Background

Elementary : Aglanot Integrated School
Aglanot, Dumarao, Capiz
2010-2016
Secondary : Ramon A. Benjamin Sr. National
High School
Dacuton Dumarao Capiz
2016-2022
Tertiary : Passi City College
City of Passi, Iloilo
Degree : Bachelor of Science in Information
Technology